What Connects Local Housing Markets? Regulation, Migration and the National Cycle

Oğuz Yurtoğlu*

First draft: July 31, 2026 This version: October 1, 2026

Abstract

Housing cannot be traded across regions, yet U.S. house prices have become far more synchronized. I estimate directed networks of house-price forecast-error dependence for the 48 contiguous states and the 100 largest metropolitan areas, 1985–2025, with and without the common factor of each ten-year window. The rise in connectedness is almost entirely that factor, which has grown from a fifth to two-thirds of the variance of state price growth, and dependence beyond it has not grown. Beyond the national cycle, land-use regulation lines up with how much dependence runs through a market and migration with where it comes from. A one-standard-deviation higher regulation index is associated with 7.5 log points more of other states' forecast-error variance attributed to a state. Regulated metropolitan areas take more of their variance from other areas where their own labor demand is rising, as a binding supply constraint implies, and which out-of-state areas that dependence comes from lines up with migration links.

JEL: R31, R23, C32, C33 **Keywords:** house prices, connectedness, migration networks, spatial equilibrium, housing supply

*Koç University, Department of Economics. Email: oyurtoglu22@ku.edu.tr. Preliminary draft, comments welcome. Replication code is available from the author.

1 Introduction

Housing is the textbook nontraded good. A house in Phoenix cannot be shipped to San Diego, and each market's prices are set by its own buyers and its own supply of homes. Yet U.S. house prices have become far more synchronized. In the ten-year windows ending in the late 1980s a single common factor accounts for about a fifth of the variance of a typical state's monthly house-price growth. In those ending since 2010 it accounts for two-thirds. Factor and comovement studies of state and metropolitan prices document the same rise (Del Negro and Otrok, 2007; Kallberg et al., 2014; Cotter et al., 2015; Fairchild et al., 2015; Tidwell et al., 2023). A common factor of that size can hide very different structures underneath it. Markets may still be tied to particular other markets, through the households who move between them, or they may answer to one national cycle and little else. This paper asks what connects local housing markets once their common cycle is taken out.

I measure those connections as a directed network of forecast-error dependence and ask which characteristics of a market they line up with. For every pair of markets the network records how much of the target's house-price forecast-error variance is attributed to price surprises in the source. Each market therefore has two sides. Its receiving side is its *external connectedness* (FROM), the share of its own forecast-error variance attributed to other markets. Its sending side is its TO connectedness, the share of other markets' forecast-error variance attributed to it. How its external connectedness divides across particular sources is its *allocation*. The national cycle complicates all three. A market whose price moves closely with a common factor shares variation with every other market, and the decomposition credits that shared variation to both sides of its position. I therefore estimate every network twice. The *raw* network is estimated on house-price growth as observed. The *factor-adjusted* network is estimated on growth from which each window's first principal component has been removed, and it measures dependence beyond the national cycle.

The networks are Diebold–Yilmaz connectedness networks estimated in rolling 120-month windows (Diebold and Yilmaz, 2012, 2014), from an elastic-net vector autoregression in monthly house-price growth whose generalized forecast-error variance decomposition at a twelve-month horizon gives the connectedness matrix. I estimate one system for the 48 contiguous states and one for the 100 largest metropolitan areas, which are closer to the markets in which supply constraints bind. No migration, distance or regulation measure enters the autoregression; those variables meet the estimated networks in a second stage, on the annual networks from 1985 through 2025 for a market's position and from 2005 for the allocation of its dependence, because migration exposure is pooled from IRS flows observed before each window begins.

The raw networks describe a strongly and increasingly connected system. Systemwide connectedness, the average share of a state's forecast-error variance attributed to other states, rises from about 68 percent in the late 1980s to about 90 percent since 2020. States differ far more on the sending side than on the receiving side, and bilateral dependence falls with distance, by less over time. Almost all of this is the common factor. With each window's first principal component removed,

systemwide connectedness stays near 65 percent throughout, the distance gradient no longer flattens, and the spread of TO connectedness across states falls by about two-fifths. Dependence orthogonal to the national cycle has not grown. The purge cannot say whether the factor's growth reflects common shocks or propagation among markets. What remains beyond it is uneven across markets and still falls with distance, and the question becomes what it lines up with.

Housing supply is the first place to look. Where building responds little, a demand shift raises prices rather than the stock (Glaeser et al., 2008; Saiz, 2010; Paciorek, 2013). The location-choice model of Schubert (2024) organizes the comparisons. Households choose among destinations, so a price rise in one market redirects demand toward the others its households consider. The weight a target's price places on a source's price is then the product of two terms. A supply multiplier sets how much the target's price moves with other markets, and it rises as the target's housing supply becomes less elastic. Migration exposure sets which markets it moves with, and it is built from where the target's residents came from and where each origin's residents move. The model has two predictions. A constrained target should take more of its dependence from elsewhere. Migration should order a target's sources while its regulation should not, because the multiplier scales every weight in the target's row and cancels from the shares. Durable housing sharpens the first prediction. A supply constraint binds when demand rises, while durable housing limits the response to falling demand in every market (Glaeser and Gyourko, 2005), so a constrained target should take more of its dependence from elsewhere where its own demand rises and not where it falls. A constraint may also put a market's own demand shocks into its price, so that other markets respond to it. The model does not derive this sending side, and I read it informally.

Beyond the national cycle, more of other states' forecast uncertainty is traced to tightly regulated states. A one-standard-deviation higher Wharton Residential Land Use Regulatory Index (WRLURI, 2006 survey) is associated with 7.5 log points more TO connectedness in the factor-adjusted network, with a 95-percent interval of $[2.7, 12.7]$, against 10.5 in the raw network. A state at the 90th percentile of regulation accounts for about 20 percent more of other states' forecast-error variance than one at the 10th, and regulated states sit closer to being net sources. Size and natural amenities are associated with a larger sending side in the raw network but not in the factor-adjusted one. Regulation is associated with it in both, under alternative treatments of the autoregression's own lags, within Census regions, with the 2018 vintage of the index, before and after 2005, and without any single state or region. It is weakest in the windows since 2020, when the national cycle leaves least room for differences between states, and it survives removing two common factors, and the lagged part of the first, but not the larger number of components an information criterion selects, so part of it may be regulated states' position within regional clusters.

The receiving side separates the national cycle from a binding constraint. In the raw network regulated states take more of their forecast-error variance from elsewhere, partly because their prices track the national factor more closely. The first principal component accounts for 8 percentage points more of their growth variance per standard deviation of the index, which is what one expects where national conditions show up in prices rather than in building (Glaeser et al., 2008; Guren

et al., 2021). Whether an average receiving-side gradient remains beyond the cycle depends on how the autoregression shrinks each state's own lags, and I leave it open. The timing of demand gives a sharper test. Across the 100 metropolitan areas I measure demand without the price data, as labor demand predicted from each area's industry mix (Bartik, 1991). Beyond the national cycle the regulation gradient of external connectedness rises with that shock, by 0.78 log points per standard deviation (0.22). Larger own shocks are the alternative for regulated places that are also productive and high-amenity (Davidoff, 2016; Diamond, 2016). They would lower external connectedness whatever their sign, so the slopes above and below the year's average shock would sum to zero. They sum to 1.42 (0.43), with a slope of 1.24 (0.37) above the average and 0.19 (0.41) below it. The interaction keeps its sign within areas over time and, with less precision, on an apportioned node set that gives each of the 48 states at least one of 100 areas. It rests on regulation. The share of land made unavailable by steep slopes, the steep land share, shows no such ordering.

Migration lines up with which markets a market's dependence comes from. I build migration exposure from IRS flows measured before each window begins, as the model defines it for the states and from flows among movers only for the metropolitan areas. Among metropolitan areas in different states, beyond the national cycle and with geography and source-by-year and target-by-year effects held fixed, a pair whose exposure is one percentage point above an equal split carries 0.07 points more of the target's external connectedness, about 9 percent of the average share of such pairs. The direct part of exposure carries the association, the estimate loses precision when nearby pairs are dropped, and the target's regulation does not change the ordering, as the model predicts. Across states migration adds nothing to geography. At that scale exposure is close to a gravity function of distance, contiguity and region, and once geography and source-by-year effects are held fixed together the association disappears.

These results connect two regularities about supply-constrained housing markets, that their prices move more with common cycles (Guren et al., 2021; Schubert, 2024; Tidwell et al., 2023) and that price shocks in them are followed by larger movements elsewhere (Vansteenkiste, 2007; Miao et al., 2011; Chiang and Tsai, 2016; Tsai and Lin, 2019; Roy Chowdhury et al., 2023), both resting on the finding that inelastic supply amplifies local booms and busts (Glaeser et al., 2008; Huang and Tang, 2012; Paciorek, 2013; Cun and Pesaran, 2022). The designs behind them do not separate a directed dependence from a shared cycle, and the distinction turns out to matter. The raw receiving side of constrained markets is partly the common cycle and depends on how the network treats own lags. Their sending side survives its removal in every own-lag network. Most other evidence on what links housing markets either relates market characteristics to symmetric comovement (Kim and Rous, 2012; Cohen et al., 2024; Hernández-Murillo et al., 2017) or fixes the links, through banking integration, economic proximity, social ties, corporate networks, contiguity or migration, and estimates how strongly prices move along them (Landier et al., 2017; Zhu et al., 2013; Bailey et al., 2018; Song, 2025; Sinai and Souleles, 2013; Cohen et al., 2016; Holly et al., 2010; Brady, 2014). Schubert (2024) is the sharpest case: his instrument, wage shocks in other cities interacted with their land constraints, identifies how strongly a destination's prices respond along

migration links fixed at a 1990–1995 baseline. Diebold–Yilmaz applications to housing estimate directed structure without imposing it, and most map where dependence is high or how its level moves with the economy rather than what it lines up with across markets (Antonakakis et al., 2018; Tsai and Lin, 2019; Payne and Sun, 2023; Gabauer et al., 2024; Antonakakis et al., 2021; Lee and Ma, 2025); closer to this paper, estimated spillovers among U.S. metropolitan areas have been related to migration, distance and industrial similarity (Malone, 2017; Wu et al., 2017), and net positions elsewhere to administrative status and size (Yang et al., 2018; Prüser and Schmidt, 2021). My raw network agrees with Gabauer et al. (2024) that the Northeast is a net receiver and New York the largest net receiver within it. This paper estimates the network from prices alone, removes the common factor before measuring the connections, as spatial and forecast-error network studies have done (Bailey et al., 2016; Barigozzi and Hallin, 2017; Huang et al., 2024), and relates both sides of each market's position and the allocation of its dependence to regulation, labor demand and migration. Where Schubert's design fixes the links to identify a response, this one measures how supply and migration align with the network and shows which alignments survive the national cycle, and it does not buy the identification his instrument buys. Every result here is an association with measured forecast dependence.

Read together, the results give regulation and migration the different roles the model gives supply and migration. Regulation lines up with how much dependence runs through a market, on the sending side across states in every own-lag network and on the receiving side across metropolitan areas where a market's own demand makes its constraint bind. Migration lines up with which markets that dependence comes from among metropolitan areas in different states. The associations follow the model's allocation prediction and, once durable housing is added, the timing its supply multiplier implies. They do not identify causal pass-through, the regulation index describes a market's land-use regime together with its demand correlates and its state-level ratings, the one-factor purge leaves regional common structure in place, and every interval reported here conditions on the estimated networks.

2 Data and Measurement

I use the seasonally adjusted monthly Freddie Mac House Price Index (FMHPI) for the 48 contiguous states from January 1975 through December 2025. The index is based on repeat transactions for single-family homes financed with conforming mortgages, those purchased by Freddie Mac or Fannie Mae. It provides long and consistent state coverage but does not represent transactions outside that segment. Alaska and Hawaii are excluded since they are detached from the mainland. States cover the whole nation but do not represent single housing markets, so a second sample of the 100 largest core-based statistical areas (CBSAs) by 2000 Census population is used as well. Small markets' repeat-sales indexes rest on few transactions, so their monthly growth is noisier, and with more and noisier series the penalty would choose among many nearly collinear lags; a size rule keeps the retained links from being close to arbitrary. The 100 areas hold 63 percent of the national

housing stock, and their price growth weighted by housing stock tracks the national index with a correlation of 0.994 over 1976–2025. The node set is defined by size alone, so it overrepresents the large states and leaves some small states without a sample metropolitan area. The same price series, transformation and estimation apply to this system.[1] The metropolitan system allows asking whether the state-level associations hold across metropolitan areas and, where a state contains several sample areas, within states.

2.1 Vector autoregression

Measuring dependence across the regional markets requires a model of their joint dynamics, and dependence need not be constant over five decades. I therefore estimate the network in rolling windows, as in the dynamic connectedness analysis of Diebold and Yilmaz (2014), so that every endpoint has its own network summarizing the months before it. The benchmark window is 120 months. When the text refers to connectedness in a year, it means the annual network, estimated on the 120-month window that ends with that year, and the window's last month is the year's endpoint. Because each network summarizes the ten years before its endpoint, movements in connectedness need not reflect developments in the endpoint month alone. The figures and the descriptive statistics use every month's window, and the regressions use the annual networks. If $P_{i,s}$ denotes the index for market i in month s, monthly house-price growth is

$$y_{i,s} = \Delta \log P_{i,s}, \tag{1}$$

where log is the natural logarithm. Collect the growth rates of month s in the $N \times 1$ vector $\mathbf{y}_s$, where N is 48 for the states and 100 for the metropolitan areas. Within each estimation window, I use a vector autoregression (VAR) of order K,

$$\mathbf{y}_s = A_0 + \sum_{\ell=1}^{K} A_\ell \mathbf{y}_{s-\ell} + \mathbf{u}_s, \qquad E(\mathbf{u}_s) = 0, \quad E(\mathbf{u}_s \mathbf{u}_s') = \Sigma. \tag{2}$$

Here A_0 is an intercept vector, A_ℓ contains the lag-ℓ relationships among markets, and $\mathbf{u}_s$ is the reduced-form disturbance. Following the modern Diebold–Yilmaz formulation, the VAR is a flexible dynamic approximation used to construct forecast-error variance shares (Buchwalter et al., 2026). Its innovations are disturbances in this fitted model, and their covariance Σ is estimated from the fitted residuals. Because the number of lagged coefficients is large relative to a rolling sample, I estimate each equation with an elastic-net penalty, which shrinks and selects lag coefficients and leaves the intercept unpenalized, following Demirer et al. (2018). No migration weight, distance or regulation measure enters the VAR or influences which links the estimator retains, so every comparison that follows sets an object estimated from prices alone beside measures built independently of it.

The benchmark is a VAR(2) estimated over rolling 120-month windows, with a mixing weight of 0.5 giving equal weight to the lasso and ridge penalty components. Penalties are selected equation

[1] See Online Appendix Section OA.6.

by equation by hv-block cross-validation (Burman et al., 1994; Racine, 2000). The 120 months of a window are cut into twelve contiguous blocks of ten, each block in turn is held out, and the months on either side of it are dropped from the training sample, as many as the largest lag in the equation, so that no training observation shares a lag window with a validation observation; the penalty is the one that minimizes validation loss. The buffer matters because the fitted residuals are serially correlated, which is the condition under which Bergmeir et al. (2018) show plain K-fold validation to fail for autoregressions. The own lags of each equation carry the same penalty as the cross lags, and none of the lag order, the minimum-loss penalty rule and the equal treatment of own and cross lags optimizes residual whiteness.[2]

Forecast-error variance shares are computed from the moving-average representation of the fitted VAR, which converges only when the system is stable, meaning that every eigenvalue of its companion matrix lies inside the unit circle (Lütkepohl, 2005). Every statistic the paper reports therefore uses stable endpoints only. All four benchmark networks, raw and factor-adjusted for the states and for the metropolitan areas, are dynamically stable in every year from 1985 through 2025, and the largest companion-matrix modulus of the raw state network is 0.978. The monthly figures also draw the endpoints at which a network is not stable, six in the raw state network (March through June 2004 and March and April 2005), two in the factor-adjusted state network (March and April 2004), four in the raw metropolitan network (March through June 2004) and none in the factor-adjusted metropolitan network. The windows ending in those months cover most of the run-up to the housing boom's peak, and the connectedness drawn for them lies close to that of the neighboring months, but no reported statistic uses them.

2.2 GFEVD and connectedness measures

The fitted VAR supplies the dynamics and innovation covariance needed to measure dependence. For a stable window with fitted mean $\bar{\mathbf{y}}$, its moving-average representation is $\mathbf{y}_s - \bar{\mathbf{y}} = \sum_{h=0}^{\infty} \Theta_h \mathbf{u}_{s-h}$, where $\Theta_0 = I$ and the remaining Θ_h matrices follow from the VAR coefficients. The generalized H-step forecast-error variance decomposition (GFEVD) assigns part of a target market's forecast uncertainty to an innovation in a source market (Koop et al., 1996; Pesaran and Shin, 1998). With target i in the row, source j in the column and t indexing the rolling endpoint, the unnormalized share is

$$\theta^{g}_{ij,t}(H) = \frac{\sigma^{-1}_{jj,t} \sum_{h=0}^{H-1} \left(\mathbf{e}_i' \Theta_{h,t} \Sigma_t \mathbf{e}_j\right)^2}{\sum_{h=0}^{H-1} \mathbf{e}_i' \Theta_{h,t} \Sigma_t \Theta_{h,t}' \mathbf{e}_i}. \tag{3}$$

[2]Appendix B.1 gives the reasons behind each choice and the diagnostics of the fitted system. Section 6.1 re-estimates the state network with the own lags exempt from the penalty, at the benchmark lag order and with eight own lags, and those two alternatives accompany the state results. The metropolitan results rest on the benchmark network alone, because the metropolitan system with its own lags exempt from the penalty is unstable at many endpoints (Online Appendix Section OA.3.4).

The vector $\mathbf{e}_i$ selects market i, and $\sigma_{jj,t}$ is source j's innovation variance in Σ_t. The generalized decomposition leaves innovations correlated, so the unnormalized shares need not sum to one. Following Diebold and Yilmaz (2012, 2014), I normalize each target row to 100,

$$C_{ij,t}(H) = 100\frac{\theta^g_{ij,t}(H)}{\sum_{k=1}^{N}\theta^g_{ik,t}(H)}, \qquad \sum_{j=1}^{N} C_{ij,t}(H) = 100. \tag{4}$$

Thus $C_{ij,t}$ is the percentage-point contribution assigned to source j in target i's normalized forecast-error decomposition. The forecast horizon is $H = 12$ months throughout, and I suppress H when no ambiguity arises. The shares are invariant to market ordering and measure reduced-form forecast dependence.

The row and column totals summarize different sides of this network. Target i's *external connectedness*, conventionally called FROM connectedness, is the share assigned to all other markets. I denote it by $F_{i,t}$. Source j's TO connectedness sums its contributions to other targets. NET connectedness is the difference between the two (Diebold and Yilmaz, 2012), and the Diebold–Yilmaz Connectedness Index (DYCI) averages the external row totals,

$$\begin{aligned} F_{i,t} &= \sum_{j\neq i} C_{ij,t} = 100 - C_{ii,t}, & T_{j,t} &= \sum_{i\neq j} C_{ij,t}, \\ \mathrm{NET}_{i,t} &= T_{i,t} - F_{i,t}, & \mathrm{DYCI}_t &= \frac{1}{N}\sum_i F_{i,t}. \end{aligned} \tag{5}$$

External connectedness is a market's *receiving side* and TO connectedness its *sending side*, and a market with NET below zero is a *net receiver*. NET itself cannot be logged, so the regressions use $\log \mathrm{NET}_{i,t} = \log(T_{i,t}/F_{i,t})$, which is positive for a net source and negative for a net receiver. Source *allocation* is how external connectedness divides across sources.

2.3 The national cycle and the factor-adjusted network

House-price growth across the regional markets shares a strong common component, and its weight has grown, as dynamic-factor and comovement studies of U.S. state and metropolitan prices have found (Del Negro and Otrok, 2007; Kallberg et al., 2014; Cotter et al., 2015; Fairchild et al., 2015; Tidwell et al., 2023). Because that literature finds the common factor central to regional house prices, the paper accounts for it and asks whether what the network measures changes once the factor is removed. In each 120-month window I take the first principal component of the standardized monthly growth rates. A network estimated on the raw series combines dependence that runs through the common factor with dependence among markets beyond it. Removing common factors before estimating the connections among units has precedents in spatial models of house prices (Bailey et al., 2016) and in forecast-error variance networks (Barigozzi and Hallin, 2017), and a dynamic factor model serves the same purpose in the factor-adjusted networks of Barigozzi et al. (2024). Here, within each window, each market's monthly growth is regressed on the window's first

principal component. The vector autoregression, the penalty selection and the decomposition are then applied exactly as above to the residuals.[3] Every design choice of the benchmark carries over, and the factor-adjusted cells and totals follow equations (4)–(5).

Two properties of the purge govern how it is read. It removes one factor at the same month, and the robustness checks also remove that factor's score at up to six lags, which is the dynamic one-factor representation, without changing the result it bounds.[4] Correlation among markets of the same Census region survives it, so a factor-adjusted result shows structure beyond the national cycle but cannot separate spillovers from regional common shocks. And the purge removes any within-month dependence among markets that happens to load on the first component, so the factor-adjusted network is a conservative measure of dependence beyond the cycle. Removing two factors, or the number a Bai and Ng (2002) criterion selects, is a stricter version of the same adjustment.[5] The purge also does not separate common shocks from propagation. A disturbance that reaches other markets within a month, or with a lag that the smooth repeat-sales series turn into comovement, also loads on the first component, so a growing common component can reflect propagation as well as common shocks. Every result is reported for the *raw* network and, where the text or table says so, the *factor-adjusted* one.

3 Connectedness

This section describes the networks before the regressions of Sections 4 and 5 ask what they line up with. Each figure draws the raw network and the factor-adjusted one, and each subsection reads the raw result first.

3.1 Systemwide connectedness and the national factor

Figure 1 plots the monthly DYCI of the raw state network, in grey, beside the factor-adjusted index and the national real house price index. The raw index drifts up from the high 60s in the late 1980s to about 75 in 2000 and then dips slightly during the housing boom, to 70 in late 2004. The dip is consistent with a boom that was still regionally concentrated. The first metropolitan booms occurred in northern New England and coastal California and spread from there (Ferreira and Gyourko, 2023), and in 2002 and 2003 appreciation was in double digits only in the Pacific, Middle Atlantic and New England divisions.[6] The index rises as the boom gives way to the Global

[3] Within the cross-validation, the component is estimated on the training months of each fold alone, so the held-out months do not enter the purge that is applied to them. Appendix C documents the purge and its stability from one window to the next.

[4] The in-sample common component of the generalized dynamic factor model, estimated with the fnets package of Barigozzi et al. (2024), was also tried as a purge. At a 120-month window it removes no common variation from these series, so the dynamic case is tested through the lagged scores instead.

[5] See Table 5, Panel A.

[6] The division growth rates are those of Office of Federal Housing Enterprise Oversight (2004). Credit expanded at the same time. Mortgage originations reached what was then a record $3.8 trillion in 2003 on a refinancing wave (Office of Federal Housing Enterprise Oversight, 2004), mortgage credit grew fastest in subprime ZIP codes from 2002

Financial Crisis, from 71 in early 2006 to 84 at the end of 2008, over about three years and in steps. After the COVID-19 disruption it jumps, from 79 in January 2020 to 88 by July 2020, and reaches its maximum of 93.5 in April 2025. In the Global Financial Crisis prices fell, and a rise in connectedness reads as the systemic-risk signal it is in the Diebold–Yilmaz literature. After COVID-19 prices rose almost everywhere at once, so a systemic-risk reading would be a stretch, since affordability worsened but no housing crisis followed.

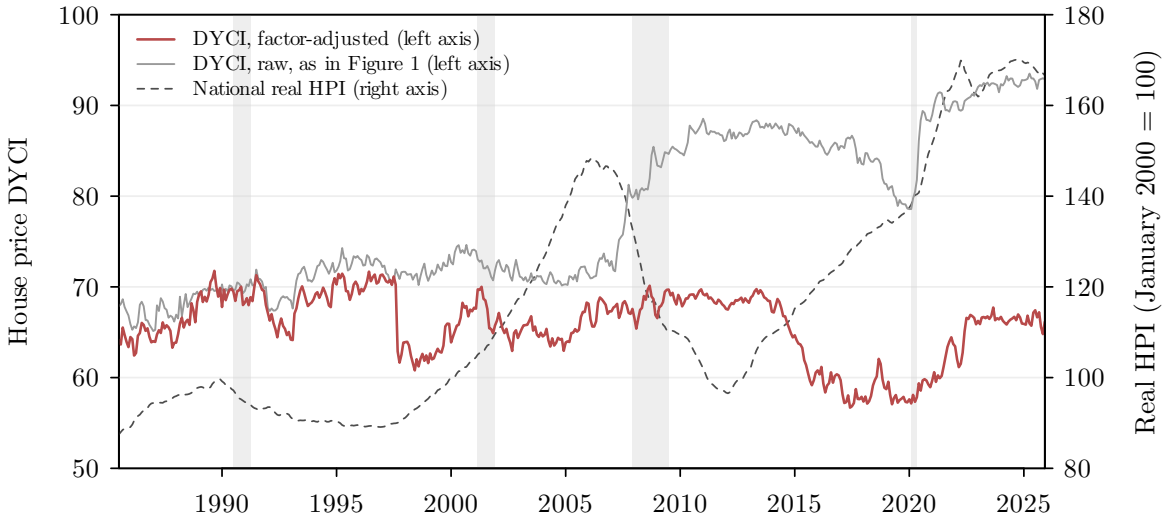


Figure 1: Systemwide house-price connectedness, raw and beyond the national factor, and the national real house price index, July 1985–December 2025

Note: The grey series is the monthly Diebold–Yilmaz Connectedness Index (DYCI) of the raw 48-state network from the 120-month rolling elastic-net VAR(2) at the benchmark specification, and the red series is the index of the factor-adjusted network, estimated after each window's first principal component of monthly growth is removed. Both are read on the left axis. The dashed series is the national real house price index, normalized to 100 in January 2000 and read on the right axis. Online Appendix Section OA.1.2 describes its construction, and its October 2025 value is missing. Every month is drawn, including those whose fitted companion matrix has an eigenvalue outside the unit circle, six in the raw network (March–June 2004 and March–April 2005) and two in the factor-adjusted one (March and April 2004), and no statistic in the text uses those windows. Shaded areas denote NBER recessions.

The same index estimated on the 100-metropolitan-area system is both higher and much steadier, 91.2 against 78.2 on average over 1985–2025, with about two-fifths of the state index's amplitude, and it turns at the same times, correlating with the state index at 0.985 in levels and 0.793 in twelve-month changes. The level gap is partly mechanical. A metropolitan target has 99 external sources where a state has 47, and part of any index of this kind is noise in the estimated innovation covariance: with no links among the series and uncorrelated innovations the index would still be about $100(N-1)/(T+N-1)$ points, 28 for 48 series and a 120-month window and 45 for 100 series, so levels are read against that floor and compared across systems of different size with care.[7]

to 2005 (Mian and Sufi, 2009), and the January 2004 federal preemption of state anti-predatory lending laws for national banks raised lending in the affected states (Di Maggio and Kermani, 2017).

[7]The floor is the index the row normalization assigns to the squared sample correlations of N independent innovations over T months; it is derived and checked by simulation in the companion paper on tuning (Yurtoğlu, 2026). The level of the index also depends on how the VAR treats each equation's own lags, but its movements largely

Removing the national factor changes the picture. The factor-adjusted index averages 65.7 against 78.2 for the raw index, and it is nearly flat. It averages 66.1 in the windows ending in 1985–1989 and 64.0 since 2020, while the raw index rises from 67.8 to 90.5, so the gap between the two grows from about 1.7 points in the late 1980s to 26.5 since 2020. The two are shares of different totals, the forecast-error variance of raw growth and of growth beyond the first component, so the gap is a difference between two indexes rather than a decomposition of one. Month by month the factor-adjusted index stays between 56.7 and 71.8, through both housing episodes in which the raw index rises, and the two indexes are weakly negatively correlated, at −0.20. The rise in systemwide connectedness since the 2000s is therefore almost entirely a rise in the component common to all markets, and dependence orthogonal to it has not grown. The metropolitan system's own first component is purged in the same way across the 100 areas, and its factor-adjusted index averages 86.2, against 91.2 raw, stays between 81.8 and 89.1 in every month and, like the state one, does not share the rise of the raw index.[8]

3.2 Directional positions

The systemwide index averages over targets, and it hides how unevenly the dependence is spread across them. Because every off-diagonal cell enters one row total and one column total, external connectedness and TO connectedness have the same cross-state mean in every month, 78.2 percentage points on average over 1985–2025 in the raw network. Their dispersion is very different. Pooled across states and months, external connectedness has a standard deviation of 12.6 percentage points and lies between 59.7 and 92.7 from the 10th to the 90th percentile. TO connectedness has a standard deviation of 45.9 and lies between 32.4 and 135.2, so a state at the 90th percentile accounts for about four times as much of other states' forecast-error variance as one at the 10th. States differ far more in how much of the system's dependence is attributed to them than in how much of their own they take from elsewhere, and NET runs from −51.0 at the 10th percentile to 56.4 at the 90th. The metropolitan areas show the same asymmetry more sharply: pooled across areas and months, external connectedness has a standard deviation of 6.0 percentage points and TO connectedness of 42.7.

Averaged by Census region (Figure 2, panel a), the West is a net source in the raw network, by 8.0 percentage points over 1985–2025 and 17.4 over 2005–2025. The South is close to balance, at 2.6, and the Midwest and the Northeast are net receivers, at −6.2 each.[9] Within the Northeast, New York is the largest net receiver, below zero in 91 percent of the windows and in all but one of those ending from 2005, as in the state network of Gabauer et al. (2024).[10] The most regulated states of

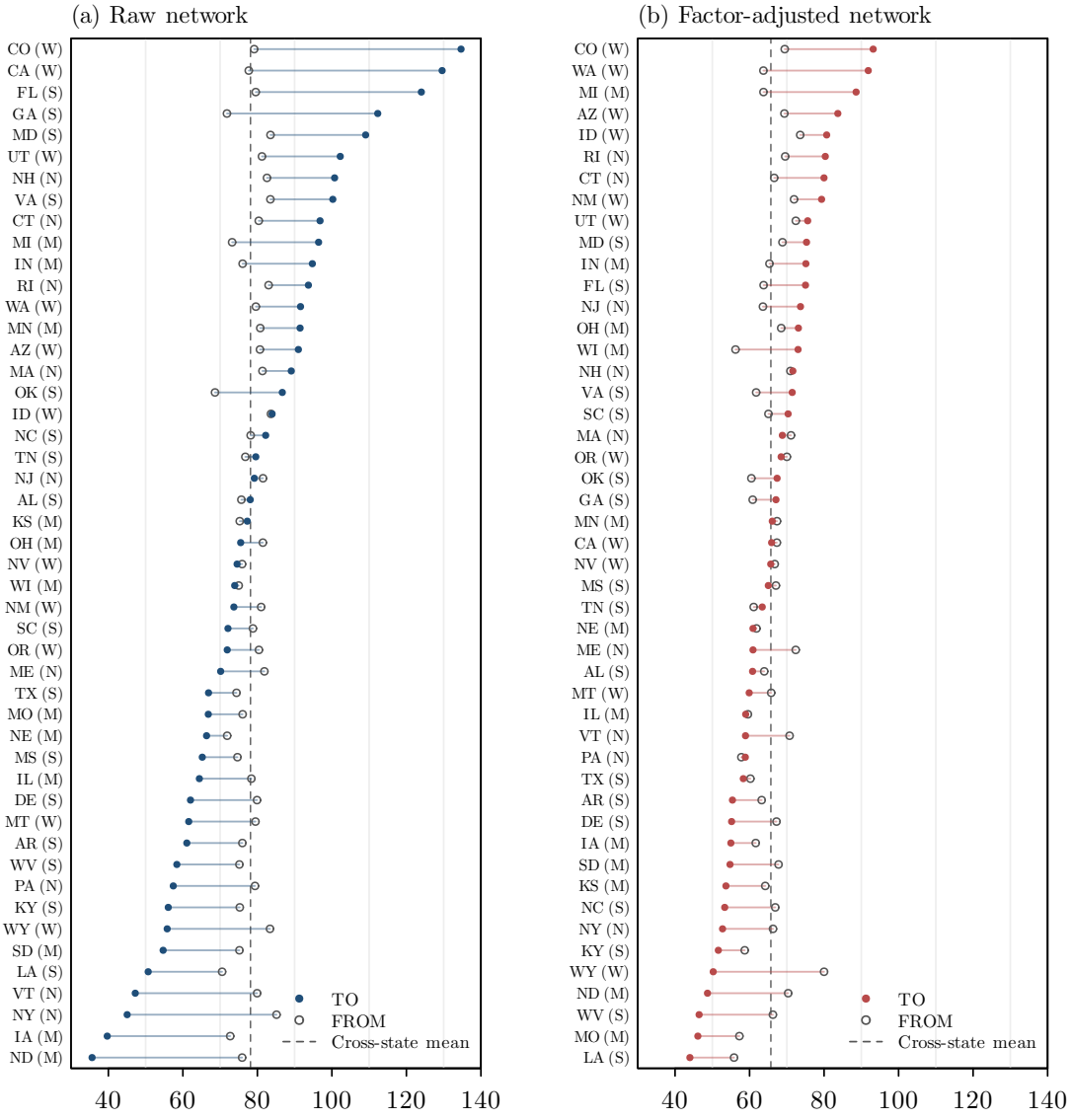
do not. With the own lags exempt from the penalty the index averages 82.9 against 87.5 for the benchmark since 2010, and over 1985–2025 the two correlate at 0.985 in levels. With eight own lags it is lower still, at 72.2, and its fitted system is explosive in two-thirds of the windows ending in the 2000s, so it is not compared. See Appendix B.1.

[8] See Online Appendix Figure OA.1, which draws the metropolitan index raw and factor-adjusted, and Online Appendix Section OA.6 for the metropolitan networks.

[9] See Online Appendix Table OA.24.

[10] Their network uses real year-on-year growth for 50 states and the District of Columbia, against nominal monthly growth for the 48 contiguous states here.

the region, Massachusetts, Rhode Island and New Hampshire, are net sources on average. Across all 48 states, California, Georgia, Colorado, Florida and Oklahoma have the highest mean log NET, and North Dakota, New York, Iowa, Vermont and Wyoming the lowest. These are positions in the raw network, and part of their spread is the national cycle, which a market that tracks it closely shares with every other market.



Mean 1985–2025, percentage points of forecast-error variance

Figure 2: Directional positions of the 48 states, raw and factor-adjusted networks, 1985–2025

Note: Each state's mean TO connectedness (filled) and external connectedness, FROM (open), over 1985–2025 in the benchmark network, in percentage points of forecast-error variance, in the raw network (a) and the factor-adjusted network (b). The segment joining the two is the state's mean NET. In each panel states are ordered by their mean TO connectedness in that network, highest at the top. The two panels share one horizontal scale. The dashed line is the cross-state mean, which FROM and TO share by construction, 78.2 in the raw network and 65.7 in the factor-adjusted one. The letter after each state is its Census region (Northeast, Midwest, South, West).

The purge narrows the directional positions, and it narrows the sending side most. Mean external connectedness falls from 78.2 to 65.7 percentage points, so the common factor accounts for about 12 points of each state's external connectedness, while its standard deviation pooled across states and months is almost unchanged, 12.2 against 12.6. The standard deviation of TO connectedness falls from 45.9 to 27.0, and that of NET from 46.9 to 29.0, so the purge narrows the spread in net position by about two-fifths. A market that tracks the common factor more closely has more of its own row attributed to other markets and a larger share of their rows, and the purge removes that difference among markets along with the factor's level. At the metropolitan scale the purge narrows the sending side by less, the standard deviation of TO falling from 42.7 to 33.2 and that of NET from 41.9 to 30.3, reductions of 22 and 28 percent against 41 and 38 percent across the states. The ordering of states changes but survives in part (Figure 2, panel b). Across the 48 states, the rank correlation between a state's mean position in the raw and the factor-adjusted network is 0.77 for TO, 0.63 for external connectedness and 0.74 for log NET. Beyond the national cycle the largest net sources by mean log NET are Washington, Michigan, Colorado, Connecticut and New Jersey, and the largest net receivers Wyoming, North Dakota, West Virginia, New York and South Dakota.

3.3 The distance gradient

Pairwise connectedness should have a spatial structure, since nearby markets are more closely linked. Figure 3 reports its spatial gradient estimated month by month by Poisson pseudo-maximum likelihood (PPML), for each window t,

$$E\left[C_{ij,t} \mid i, j\right] = \exp\left(\lambda_{i,t}^{T} + \lambda_{j,t}^{S} - \beta_t \log d_{ij}\right), \qquad i \neq j, \tag{6}$$

where d_{ij} is the distance between target i and source j and $\lambda_{i,t}^{T}$ and $\lambda_{j,t}^{S}$ are target and source fixed effects. The log link keeps the zero-valued cells and makes β_t an inverse-distance elasticity. The target effects absorb the level of each row but not its weight, so a target with a larger external total weighs more in the estimate; dividing each row by its external total gives every target the same weight, and the gradients estimated on those shares correlate with the ones drawn here at 0.996 or more, at both scales and in both networks.

In the raw state network the inverse-distance elasticity is positive at every point at which the system is dynamically stable, and at every one of them the 95-percent interval excludes zero. Its mean is 0.278. Geographic proximity is therefore associated with stronger bilateral connectedness throughout the period, but by a diminishing amount, as the mean elasticity falls from 0.410 in the 1990s to 0.117 during 2020–2025. Re-estimating the monthly regressions jointly, with source and target effects specific to each window, leaves every point estimate unchanged and adds one covariance matrix: against the 1990s mean of 0.410, the mean over 2015–2025 is 0.109, a fall of 0.301 with a two-way clustered standard error of 0.045.[11] Across the 100 metropolitan areas the

[11] See Online Appendix Section OA.2.1. The network with unpenalized own lags gives the same picture along a slightly flatter gradient, with a mean of 0.264 and the same decline, from 0.348 in the 1990s to 0.125 during 2020–2025.

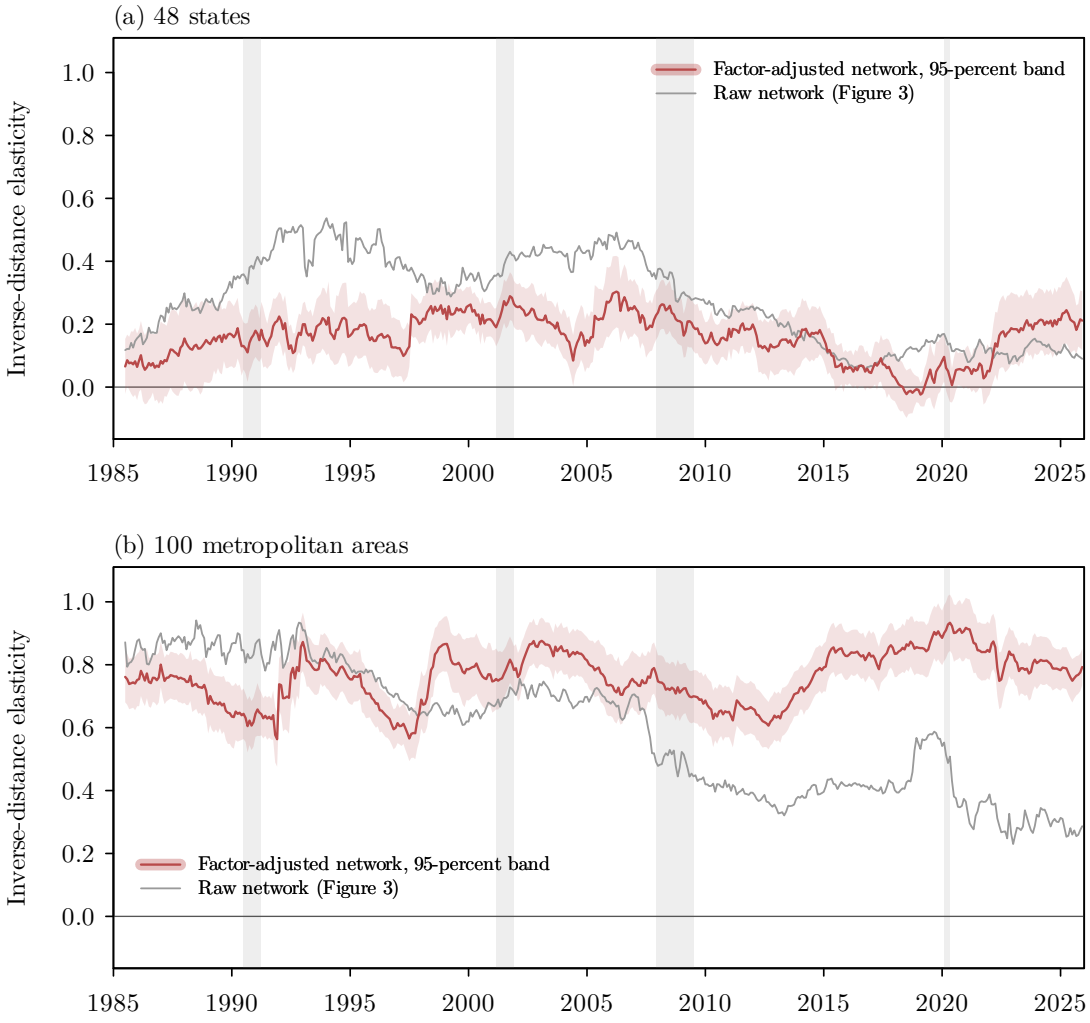


Figure 3: The distance gradient in bilateral connectedness, raw and factor-adjusted networks, July 1985–December 2025

Note: Each line reports the coefficient on negative log distance from separate Poisson pseudo-maximum-likelihood regressions of the nonnegative $H = 12$ bilateral connectedness cell, one regression per month, in the raw (grey) and the factor-adjusted (red, with its 95-percent band) benchmark network. Panel (a), 48 states, uses interstate distances and panel (b), 100 metropolitan areas, distances between CBSA internal points. Every regression includes source-node and target-node fixed effects. The band is a 95-percent confidence interval based on two-way source and target clustering, and grey vertical bands denote NBER recessions. Every month is drawn. In the windows whose fitted VAR is not stable, six raw and two factor-adjusted state windows and four raw metropolitan ones, the gradient is estimated for the figure only, and no statistic in the text uses it.

gradient is about twice as steep, averaging 0.59, positive with an interval excluding zero in every month, and it declines in the same way, from 0.77 in the 1990s to 0.33 during 2020–2025, correlating 0.69 with the state gradient month by month. The decline is a fact about house prices rather than an artifact of how the network is estimated. The correlations of monthly price growth among state pairs, computed over the same windows without any VAR, line up with proximity most strongly in the mid-1990s and least in the mid-2010s, as the estimated gradient does, and the two correlate at 0.68 month by month.[12]

The factor-adjusted networks change the spatial description. The factor-adjusted state gradient averages 0.158, against 0.278 raw in the same months. It is positive in 98 percent of the months, with an interval excluding zero in 80 percent, and it does not decline: it averages 0.11 in the windows ending in 1985–1989 and 0.14 since 2020.[13] The factor-adjusted metropolitan gradient averages 0.76, against 0.59 raw, and rises slightly, from 0.73 in the late 1980s to 0.83 since 2020, whereas the raw gradient falls from 0.86 to 0.33. The decline of the distance gradient in the raw networks is therefore associated with the growth of the national cycle, which raises the dependence of distant and nearby markets alike. Once the national factor is removed, the gradient shows no decline across states and a slight rise across metropolitan areas, and the two series differ materially, the raw estimate lying outside the factor-adjusted interval in about two-thirds of the months at the state scale and four-fifths at the metropolitan one.

Two limits of the purge, that it removes one factor and that it cannot separate common shocks from propagation among markets, bound how far the factor-adjusted networks can be read as structure beyond the national cycle. What the comparison does show is that dependence orthogonal to the national cycle has not grown, that markets still differ in how much of it is attributed to them, and that it still falls with distance. The next sections ask which characteristics of a market line up with those differences, in the raw network and beyond the national cycle.

4 Housing Supply, Migration and the Empirical Design

How much of a demand shift shows up in a market's price depends on its housing supply. Where building responds little, higher demand raises prices rather than the stock, so the price of a supply-constrained market moves with the demand of the households who might live there (Glaeser et al., 2008; Saiz, 2010; Paciorek, 2013). The premise is not settled for recent decades. Across U.S. cities, measured constraints do not change how price and quantity growth respond to income growth (Louie et al., 2025), and the comparisons below ask a different question, how forecast-error dependence and

[12]The check computes, in each window and over the same 120 months, the correlation of monthly price growth for all 1,128 state pairs and asks how strongly those correlations line up with proximity. The periods of strongest comovement are mostly ones in which proximity explains least about which states move together, although the recovery of the correlation-based measure after 2020 qualifies that statement for the most recent windows. See Online Appendix Figure OA.2.

[13]Across the annual networks its linear trend, −0.001 a year, is not distinguishable from zero, against −0.008 raw, with a Newey–West interval with nine lags, because windows up to nine years apart share months. At the metropolitan scale the two annual trends have opposite signs, and the factor-adjusted interval excludes zero.

its timing line up with regulation. Those households come from other markets, and migration says from which. They can connect two housing markets even when they do not move directly between them. Suppose households from the same origin commonly consider two destinations. When housing in one destination becomes more expensive, some may choose the other, and the destinations are then linked through competition for a common origin population. The location-choice model in Schubert (2024) formalizes both parts. Migration determines which markets a target's price responds to, and the target's housing-supply elasticity determines how strongly it responds.[14]

A household initially located in k chooses among destinations by comparing amenities, wages, housing costs and moving costs. Let $\mu_t^{k\to j}$ denote the share of origin-k households choosing destination j, and let $\varphi_t^{i\leftarrow k}$ denote the share of target i's feeder population supplied by origin k. Migration-network exposure from source j to target i is

$$\psi_{ij,t} = \sum_{k=1}^{N} \varphi_t^{i\leftarrow k}\mu_t^{k\to j}, \qquad \sum_{j=1}^{N}\psi_{ij,t} = 1. \tag{7}$$

In the data, $\psi_{ij,t}$ is built from the pooled IRS flows of the years $t-15$ to $t-11$.[15] Exposure is high when origins important to target i also assign a large share of their destination choices to source j. It incorporates all shared origins and therefore contains more information than the direct flow between i and j. For $j \neq i$ it splits exactly into two direct terms and a common-origin term,

$$\psi_{ij,t} = \varphi_t^{i\leftarrow i}\mu_t^{i\to j} + \varphi_t^{i\leftarrow j}\mu_t^{j\to j} + \sum_{k\neq i,j}\varphi_t^{i\leftarrow k}\mu_t^{k\to j}, \tag{8}$$

residents of i who move to j, residents of i who came from j weighted by the share of j's residents who stay, and the origins the two markets share. With stayers kept on the diagonal of the state flow matrix, as the model requires, the two direct terms dominate: across states the common-origin term averages 1.3 percent of external exposure, so exposure is, to a close approximation, direct migration in both directions weighted by stayer shares, while in the metropolitan flow matrices, which hold movers only, it is 39 percent.[16]

The diagonal $\psi_{ii,t}$ is the target's own exposure, and $o_{i,t} = 1 - \psi_{ii,t}$ measures its external migration openness. Conditional on choosing an external destination, source j's migration share is $\widetilde{\psi}_{ij,t} = \psi_{ij,t}/o_{i,t}$. In the model, the weight $W_{ij,t}$ that target i's price places on source j's price follows from the target's inverse housing-supply elasticity η_i^H and two common parameters, the housing expenditure share α and the dispersion ν of households' location tastes. With $a_i = (\alpha/\nu)\eta_i^H$,

$$W_{ij,t} = \rho_{i,t}\psi_{ij,t}, \qquad \rho_{i,t} = \frac{a_i}{1 + a_i o_{i,t}}, \tag{9}$$

[14] Online Appendix Section OA.9 provides the derivation, and Online Appendix Section OA.9 its continuation.

[15] Online Appendix Section OA.9 writes this pre-window matrix as $\psi_{ij,t}^0$.

[16] See Online Appendix Table OA.20. Schubert (2024) builds exposure for commuting zones and drops links within 50 to 150 miles of each zone. That exclusion has no state counterpart, and at the metropolitan scale the paper uses pairs of areas in different states instead, with a radius exclusion as a check.

where $\rho_{i,t}$, the *supply multiplier*, scales every exposure weight of target i and rises with its supply constraint. Summing over sources and dividing by that sum gives the two margins the regressions follow,

$$\sum_{j \neq i} W_{ij,t} = \rho_{i,t} o_{i,t} = \frac{a_i o_{i,t}}{1 + a_i o_{i,t}} \equiv \eta^{nw}_{i,t}, \qquad \frac{W_{ij,t}}{\sum_{r \neq i} W_{ir,t}} = \widetilde{\psi}_{ij,t}. \tag{10}$$

The target's total external weight, $\eta^{nw}_{i,t}$, therefore rises when housing supply is less elastic and when the target is more open to external migration. Conditional on that total, migration shares determine how the weight is allocated across sources. Supply sets how much of a target's price moves with other markets, and migration sets which markets it moves with. These implications motivate separate regressions for a market's directional position and for the allocation of its external connectedness across sources.

The inverse supply elasticity η^H_i of the model is a constant, so neither margin depends on the direction of the target's own demand. Durable housing adds that dependence. Supply is inelastic for falling demand in every market, because housing is not torn down when demand falls, but elastic for rising demand where building is unconstrained (Glaeser and Gyourko, 2005). The difference in effective η^H_i between constrained and unconstrained targets, and with it the difference in their total external weight, should therefore appear where a target's own demand rises and not where it falls. This timing implication combines the model with durable housing and is not one of the model's comparative statics.

The theoretical weights and the GFEVD cells are not identical. Innovation covariance, indirect paths, the forecast horizon and row normalization also shape measured connectedness, and the measured forecast-error share is far larger than the model's direct external weight, whose magnitude depends on preference and supply parameters as well as on openness. I therefore use the model to organize the empirical comparisons without restricting the VAR by migration weights.[17]

4.1 Sample, timing and variables

The directional regressions use the annual networks from 1985 through 2025, the full span over which the benchmark network exists and is dynamically stable, for the states and for the metropolitan areas, since their regressors are time invariant or reach back to 1984. The allocation regressions use the annual networks from 2005 through 2025, because IRS migration flows start with the 1990 vintage and each network's exposure is built from migration observed before its ten-year price window begins, so that no month of the window can inform the weights it is compared with. IRS vintages are labeled by the first year of a two-year migration interval, so for the annual network of year t I pool vintages $t-15$ through $t-11$, whose final interval ends in $t-10$, before the

[17] In the 48-state system from 2005, mean external migration openness is 5.69 percent against mean external connectedness of 85.29 percent in the raw network. On the unconditional cell the supply multiplier does not cancel, and Online Appendix Section OA.7 shows that the interaction of exposure with the target's regulation there is an accounting identity in the receiving-side gradient and the allocation slope.

120-month window begins in $t-9$; the earliest pool, 1990–1994, supports the 2005 network.[18] The supply constraint enters as the 2006 Wharton Residential Land Use Regulatory Index (Gyourko et al., 2008), with the 2018 index (Gyourko et al., 2021) as a check, and as the share of land made unavailable by steep slopes from Lutz and Sand (2026), the steep land share from here on. Both are standardized across the markets of each system and treated as persistent characteristics of a market's land-use regime rather than as dated treatments, so the regressions compare markets. The size control is the resident population in the calendar year before the network's year, in logs, and bilateral geography consists of log distance, contiguity and a common Census-region indicator.[19] Table 1 reports the network outcomes and the market and pair characteristics of the state and the metropolitan regressions at their observational levels.

4.2 Directional position

The framework speaks to both sides of a market's position. A constrained target carries a larger inward weight, because its supply multiplier scales every exposure weight in its own row, while a market's outward weight depends on other markets' constraints and on how many of their movers weigh it as a destination. What the direct weights imply for the measured decomposition is less definite, since the shares are generalized forecast-error shares of a fitted dynamic system, and a constraint that makes a market's own demand shocks show up in its price also makes those surprises larger relative to the responses of other markets. Where a regulated market sits on the sending side is therefore an empirical question. Write R_i for standardized WRLURI and $Y_{i,t}$ for one of $\log T_{i,t}$, the log of TO connectedness, $\log F_{i,t}$, the log of external connectedness, and $\log \mathrm{NET}_{i,t} = \log(T_{i,t}/F_{i,t})$. The model is

$$Y_{i,t} = \delta_t + \beta_R R_i + X'_{i,t}\gamma + \varepsilon_{i,t}, \tag{11}$$

where δ_t denotes year effects and $X_{i,t}$ contains the steep land share, lagged log population and, for the states, log average distance. I estimate it by least squares and report $100\beta_R$, the difference in the outcome in log points per standard deviation of the regressor, approximately a percent difference. Because $\log \mathrm{NET} = \log T - \log F$ market by market, its coefficient on every covariate equals the TO coefficient minus the external-connectedness coefficient exactly. Every directional regression is estimated twice, on the raw network and on the factor-adjusted one, with the same covariates and sample, which answers the question the raw network cannot, whether a regulated market's position reflects more than heavier comovement with the national cycle. The regressors are time invariant, so

[18] Schubert (2024) uses averages from an early baseline period to form fixed migration weights. Here the averaging window advances with the rolling house-price window, and Online Appendix Table OA.19 fixes it at the 1990–1994 pool instead.

[19] The benchmark land measure retains whole metropolitan and micropolitan areas, allocates each cross-border market to states by its shares of the market's 1990 housing stock, and weights markets within each state by that stock; the primary state WRLURI is the unweighted mean across responding jurisdictions. Appendix A reports coverage, timing and aggregation details. Migration openness exists only from 2005, and entering it would confine the directional regressions to half their span (Online Appendix Section OA.4.1).

Table 1: Summary statistics for connectedness, supply constraints and migration exposure

Variable	Unit	*N*	Mean	SD	P10	Median	P90
Panel A. States, target variables, 41 annual networks 1985–2025							
External connectedness (FROM)	pp	1,968	78.057	12.600	59.284	80.335	92.587
TO connectedness	pp	1,968	78.057	46.079	32.283	66.872	136.490
Net difference, TO − FROM	pp	1,968	0.000	47.162	-51.156	-6.959	56.350
External connectedness (FROM), factor-adjusted	pp	1,968	65.575	12.437	49.334	65.363	82.268
TO connectedness, factor-adjusted	pp	1,968	65.575	26.710	37.153	60.111	99.947
Net difference, TO − FROM, factor-adjusted	pp	1,968	0.000	28.847	-34.207	-2.210	35.575
log FROM	Log	1,968	4.343	0.176	4.082	4.386	4.528
log TO	Log	1,968	4.197	0.578	3.475	4.203	4.916
log NET = log(TO/FROM)	Log	1,968	-0.146	0.608	-0.933	-0.102	0.560
Lagged log population	Log	1,968	15.142	0.993	13.696	15.262	16.347
log average interstate distance	Log	48	7.392	0.221	7.154	7.330	7.724
Steep land share	Percent	48	16.886	16.623	2.382	11.735	42.155
WRLURI 2006	Index	48	-0.128	0.711	-0.985	-0.206	0.754
Panel B. States, directed-pair variables, 21 annual networks 2005–2025							
Conditional connectedness share	pp	47,376	2.128	1.913	0.308	1.609	4.601
Conditional migration share (uncentered)	pp	47,376	2.128	3.362	0.182	0.991	5.302
log interstate distance	Log	2,256	7.211	0.705	6.257	7.311	8.052
Contiguous states	Indicator	2,256	0.095	0.293	0.000	0.000	0.000
Same Census region	Indicator	2,256	0.246	0.431	0.000	0.000	1.000
Panel C. Metropolitan areas, target variables, 41 annual networks 1985–2025							
External connectedness (FROM)	pp	4,100	91.156	5.975	83.795	92.718	96.758
TO connectedness	pp	4,100	91.156	42.386	41.816	85.838	147.377
Net difference, TO − FROM	pp	4,100	0.000	41.598	-47.622	-4.672	53.618
External connectedness (FROM), factor-adjusted	pp	4,100	86.148	7.125	76.453	87.874	93.238
TO connectedness, factor-adjusted	pp	4,100	86.148	33.242	42.023	85.295	129.561
Net difference, TO − FROM, factor-adjusted	pp	4,100	0.000	30.302	-38.856	-0.376	39.311
log FROM	Log	4,100	4.510	0.070	4.428	4.530	4.572
log TO	Log	4,100	4.396	0.508	3.733	4.452	4.993
log NET = log(TO/FROM)	Log	4,100	-0.114	0.500	-0.745	-0.053	0.454
Lagged log population	Log	4,100	14.028	0.826	13.164	13.788	15.272
Steep land share	Percent	100	15.648	17.335	0.230	9.686	44.072
WRLURI 2006	Index	100	0.087	0.668	-0.803	0.077	1.026
Panel D. Metropolitan areas, directed-pair variables, 21 annual networks 2005–2025							
Conditional connectedness share	pp	207,900	1.010	1.100	0.156	0.738	2.037
Conditional migration share (uncentered)	pp	207,900	1.010	2.486	0.037	0.312	2.370
log distance between CBSA internal points	Log	9,900	7.211	0.809	6.135	7.324	8.168
Same Census division	Indicator	9,900	0.120	0.325	0.000	0.000	1.000
Same principal state	Indicator	9,900	0.037	0.188	0.000	0.000	0.000

Note: Panels A and B cover the 48 states and Panels C and D the 100 metropolitan areas. Panels A and C cover the annual networks 1985–2025, and time-invariant characteristics are reported once for each state or area. The factor-adjusted rows come from the network estimated after each window's first principal component is removed. The land measure is the share made unavailable by steep slopes, aggregated for the states from whole metropolitan and micropolitan markets using 1990 total housing stock (Online Appendix Section OA.1.3) and measured over the CBSA polygon for the metropolitan areas. The cross-state standard deviations used to standardize WRLURI 2006 and the steep land share are 0.71054 index units and 0.16623 share units, and the cross-area standard deviations are 0.66845 and 0.17335. TO connectedness sums a market's contributions to the other 47 states or 99 areas and is not bounded by 100; NET, TO minus FROM, sums to zero across markets at every annual network, and $\log \text{NET} = \log(\text{TO}/\text{FROM})$ is positive for a net source and negative for a net receiver. Average distance is not defined for the metropolitan areas. Panels B and D cover the 21 annual networks from 2005 and report conditional shares of the raw network over directed pair-years, with fixed geography once for each directed pair. Each conditional share sums to 100 across the 47 external states or 99 external areas, so its mean is 100/47 or 100/99 by construction. Distance is measured in kilometers before taking logs, between internal points for the metropolitan areas; pp denotes percentage points.

their coefficients compare markets rather than changes within markets, and the regressions describe directional position without estimating the model's housing-supply parameter.

The metropolitan system extends the comparison in two directions. Across the 100 areas, Census-division effects absorb the regional composition that 48 states cannot separate from regulation, and one control the state system does not need enters, the share of the population within 200 kilometers that lives outside the node set.[20] Within states, the comparison uses each area's *principal state*, the state holding the largest share of its 2010 housing units, requires at least three sample areas in a state, which twelve states meet with 64 areas, and absorbs everything about a state that does not change over time with principal-state effects, or each state's shocks in every year with principal-state-by-year effects. What remains are differences among metropolitan areas of one state, of which regulation is one; coastal location, size and amenities can also differ within a state, and the design does not remove them.

4.3 Allocation across sources

The bilateral analysis asks how a target's external share is distributed across particular sources. Conditional on $F_{i,t} > 0$, define

$$\bar{C}_{ij,t} = 100\frac{C_{ij,t}}{F_{i,t}}, \qquad j \neq i, \qquad \sum_{j \neq i} \bar{C}_{ij,t} = 100, \tag{12}$$

which allocates 100 percentage points across the 47 external sources and separates *conditional source allocation* from the target's total external connectedness. The regressor is exposure expressed as a deviation from an equal split,

$$x_{ij,t} = 100\widetilde{\psi}_{ij,t} - \frac{100}{47}, \qquad \sum_{j \neq i} x_{ij,t} = 0, \tag{13}$$

called *conditional migration exposure* from here on; a negative value means a below-average share, not a negative flow. For specification q, I estimate

$$\bar{C}_{ij,t} = \beta_{M,q} x_{ij,t} + \boldsymbol{\gamma}_q' \mathbf{G}_{ij} + \lambda_q(i,j,t) + \varepsilon_{ij,t,q}, \tag{14}$$

where $\mathbf{G}_{ij}$ contains bilateral geography when included, the three geography variables and their pairwise interactions, and $\lambda_q(i,j,t)$ denotes the fixed effects of specification q. The coefficient $\beta_{M,q}$ is the difference in the conditional connectedness share associated with a one-percentage-point difference in conditional migration exposure.

Two designs carry the comparison. The first holds source-by-year and target-by-year effects fixed. Source-by-year effects absorb each source's general pull as a destination and its prominence in the

[20] Online Appendix Section OA.6.2 motivates this control. The metropolitan system is open, retaining 61 percent of migrating exemptions against essentially all for the states, so a metropolitan target's migration exposure is measured over a partial pool of movers.

network in each year, so the migration slope compares the sources of a given target and year net of how favored each source is by migrants in general. Target-by-year effects absorb nothing from the outcome or the regressor where the row constraint holds, because the conditional shares sum to 100 within each target-year and exposure is centered within target and year, and once source-by-year effects are included they absorb only a term of order 1/47; they become a substantive step only where the row constraint is broken, in the metropolitan sample of pairs in different states. The second adds bilateral geography on top, with distance, contiguity and common region allowed to depend on one another, which separates the role of geography from that of the source pull. The sequence that builds up to the two, from the pooled slope through geography and source-by-year effects, is in the Online Appendix. At metropolitan scale each target has 99 external sources and exposure is centered at 100/99, shared Census division replaces contiguity, and because pairs of areas with the same principal state share state-level shocks as well as migrants, the metropolitan regressions keep every pair and give the same-state sources of each target-year a level of their own, with the exposure and distance slopes allowed to differ for those pairs, so that the slope for pairs in different states and the same-state increment are estimated separately while the shares still sum to 100 across every target's sources.[21]

4.4 Inference

For the state system the target regressions compare 48 states, and their regressors vary across states rather than within them, so inference clusters by state and allows arbitrary dependence over time within a state. With 48 clusters, conventional critical values for cluster-robust t statistics tend to reject too often, so every p-value and star in the target tables comes from a wild cluster bootstrap-t that imposes the null hypothesis, with 9,999 draws of Webb's six-point weights, one per cluster (Cameron et al., 2008; MacKinnon and Webb, 2017; Webb, 2023), and the 95-percent intervals quoted in the text invert the same test (Roodman et al., 2019). The metropolitan target regressions cluster by metropolitan area, and the within-state columns add the same test with the twelve principal states as clusters. A difference between the raw and the factor-adjusted coefficient comes from one regression that stacks the two networks, with every regressor and fixed effect specific to its network and both observations of a state or area in one cluster. Conditional-allocation regressions use shared-node standard errors, which allow two directed observations to be dependent whenever they share a node in either role, with stars against a Student t distribution with one degree of freedom fewer than the number of nodes, and a delete-one-node jackknife provides a separate influence check. The state-cluster inference relies on independence across states, although regulation is spatially clustered, so a wild cluster bootstrap over the nine Census divisions and Conley standard errors accompany the sending-side results (Section 6.2).

[21]Two narrower designs, reported in the Online Appendix, ask whether the association has a direction: unordered-pair effects, which absorb fixed symmetric pair attributes, and the *saturated* design, which adds source-by-year and target-by-year effects to them.

Every interval and p-value reported in this paper treats the estimated network as data. The procedures above describe sampling variation in the second-stage panels while holding the connectedness shares fixed at their estimated values, and none of the estimation error in penalty selection, the elastic-net VAR coefficients, the decomposition or, for the factor-adjusted network, the extraction of each window's first principal component is carried into the reported precision. A comparison of estimators, as in Section 6, is not a sampling distribution over networks. Obtaining one would require resampling the monthly panel, repeating penalty selection, the purge and the decomposition in every window, and re-estimating the second stage on each resulting network, and whether that uncertainty would change the reported intervals is not established here. The intervals in this paper should be read as conditional statements.

5 Results

Tightly regulated states account for more of other states' forecast uncertainty, in the raw network and beyond the national cycle. Table 2 reports the directional regressions from 1985 through 2025, the raw network in columns (1)–(3) and the factor-adjusted network in columns (4)–(6). In the raw network a one-standard-deviation increase in the Wharton index is associated with 10.5 log points more TO connectedness (column 1). Once the national factor is removed it is associated with 7.5 log points more (column 4), with a cluster-robust standard error of 2.3 and a 95-percent interval of $[2.7, 12.7]$ that inverts the wild bootstrap test. The two estimates are not distinguishable from each other. Their difference is -3.0 with a standard error of 2.6, from one regression that stacks both networks and clusters each state's observations together. Removing the national factor takes about 30 percent off the raw estimate and leaves the gradient estimated more precisely than before. Regulation marks where dependence is attributed and not only how closely a state follows the cycle.

The association is large. Beyond the national cycle, a state at the 90th percentile of regulation accounts for about 20 percent more of other states' forecast-error variance than one at the 10th. Spread over the 47 other states, a standard deviation of regulation is about a tenth of a percentage point of each one's forecast-error variance. The attribution is small for any one partner and sizable in total. Against the outcome's own dispersion the association is the same in both networks, about a fifth of a standard deviation of TO per standard deviation of regulation.[22] Regulated states also sit closer to being net sources. Per standard deviation of regulation, log NET, the log ratio of TO to external connectedness, is 7.8 log points higher in the raw network and 5.8 in the factor-adjusted one (columns 3 and 6). The net position is the less precise of the two sending-side results, because its outcome is the more dispersed and it inherits the imprecision of the receiving side. Larger states account for more of other states' variance only in the raw network, with an elasticity of TO with

[22] One standard deviation of WRLURI 2006 is 0.71 index points, and the 90th and 10th percentiles lie 2.4 standard deviations apart, so the difference is $100(e^{0.075\times 2.4} - 1) \approx 20$ percent. In levels a standard deviation of regulation is 4.9 percentage points of TO (0.075×65.6, with 65.6 the factor-adjusted mean), or about 0.1 points for each of the 47 partners. The within-year cross-state standard deviation of log TO is 0.56 in the raw network and 0.39 in the factor-adjusted one, so a standard deviation of regulation is associated with 0.19 standard deviations of TO in both.

Table 2: Regulation and directional connectedness, raw and factor-adjusted networks, 1985–2025

	Raw			Factor-adjusted		
Dependent variable	log TO	log FROM	log NET	log TO	log FROM	log NET
	(1)	(2)	(3)	(4)	(5)	(6)
WRLURI 2006	10.54**	2.76***	7.78**	7.50***	1.67	5.83**
	(3.62)	(0.60)	(3.40)	(2.32)	(1.05)	(2.08)
Steep land share	1.19	0.72	0.47	−0.71	−0.19	−0.51
	(3.43)	(0.46)	(3.25)	(2.39)	(0.87)	(2.28)
log average distance	19.12	1.82	17.30	18.69	12.69**	6.00
	(16.96)	(3.43)	(15.87)	(12.94)	(5.32)	(12.06)
Lagged log population	12.06**	−0.38	12.44**	3.29	−3.89***	7.18**
	(3.92)	(0.67)	(4.15)	(2.29)	(0.89)	(2.62)
Year fixed effects	✓	✓	✓	✓	✓	✓
State-years	1,968	1,968	1,968	1,968	1,968	1,968
Within R^2	0.097	0.056	0.071	0.072	0.083	0.053

Note: 48 states, 41 annual networks 1985–2025; raw network in columns (1)–(3), factor-adjusted in (4)–(6). $\log \text{NET} = \log(\text{TO}/\text{FROM})$, so its coefficients equal the TO minus the FROM coefficients. Population is BEA state population in the year before the network's year. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Within R^2 is net of the year effects. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws). The factor-adjusted minus raw WRLURI coefficient, from one regression stacking both networks, is −3.04 (2.64) for log TO, −1.09 (0.88) for log FROM and −1.95 (3.00) for log NET.

respect to lagged population of 0.12 (0.04) raw and 0.03 (0.02) factor-adjusted, while the regulation association is distinguishable from zero in both.

I read this as a supply-side fact about how price signals travel. A constraint on building keeps a market's own demand shocks from being absorbed by new housing, so they show up in its price, and a price that moves with demand is what households in other markets respond to when they reconsider where to live (Glaeser et al., 2008; Paciorek, 2013). The premise is consistent with Cun and Pesaran (2022), in whose estimated model a California productivity shock raises local prices more and draws fewer migrants than a comparable shock in Texas, and the result points in the direction of Vansteenkiste (2007), who finds in a global VAR of 31 states that a price shock in land-scarce California raises prices across the country while one in Texas does not. The comparison here imposes no weights and fixes no shock. TO combines how large a market's price surprises are with how strongly other markets' forecast errors load on them, so the estimates do not identify how far a given shock travels. They show that more of other markets' forecast uncertainty is traced to regulated markets beyond the national cycle. The framework derives the weight that a constrained market's price places on other markets' prices, not this sending side, so the reading is an informal extension of it. The gradient survives removing two common factors, and the first component together with up to six lags of its score, but not the larger number of components the Bai and Ng (2002) criterion selects in each window, so part of it may be regulated states' position within regional clusters (Section 6.1).

The index itself has two kinds of content, and the two sides of a market's position lean on different ones. Two of its eleven components are ratings of how involved a state's legislature and courts are in land use (Foster and Summers, 2005; Gyourko et al., 2008), which vary only across states; the other nine come from the survey of local jurisdictions. Entered beside each other with the index's own weights, the state-level components carry most of the factor-adjusted TO and log NET associations and the local survey content little of them, while across states and metropolitan areas the survey content carries the receiving side and the state ratings do not. The blocks cannot be ranked with confidence outside the factor-adjusted state receiving side, and each is positively associated with TO when entered alone.[23] The point estimates place the sending side in state institutions and the receiving side in local rules. If it is the state's legal environment, the margin of regulation that matters for the sending side operates at the scale at which it is measured, the state, and not the town.

5.1 Labor demand and the two readings of regulation

The sending side cannot by itself tell a binding supply constraint from larger demand shocks. Regulated places are productive, high-amenity places. Across metropolitan areas, supply-constraint measures are correlated with a range of demand proxies (Davidoff, 2016). High-skill cities are also more regulated, and their local amenities rise with the location choices of high-skill workers (Diamond, 2016). The metropolitan system does not separate the two readings either, because it carries the sending side less securely than the state system. Across the 100 metropolitan areas, with Census-division effects and population controls, a standard deviation of regulation is associated with 6.6 log points more TO connectedness (2.6) in the raw network and 4.6 (3.0) in the factor-adjusted one, with an interval of $[-2.1, 10.6]$.[24] The average gradient beyond the national cycle is not distinguishable from zero at this scale.

The two readings differ in timing. A constraint that turns demand into price binds when demand rises. Durable housing limits the response to falling demand in every market (Glaeser and Gyourko, 2005). A supply reading therefore predicts a steeper receiving-side gradient where a market's demand is rising, and only there. Larger demand shocks alone would make the gradient rise with the size of the shock in either direction, so they predict that its slopes above and below the average sum to zero. I measure demand without the price data, as labor demand predicted by industry mix. For each market and estimation window it is a shift-share prediction of wage-bill growth over the window, built from BEA industry earnings with shares from the year before the window and national industry growth that leaves the market out (Bartik, 1991). The prediction tracks actual growth, with F statistics of 16.2 across states and 30.9 across metropolitan areas in regressions of actual on predicted growth, and regulated markets have somewhat stronger predicted demand, the correlation between the index and the prediction averaging 0.20 across metropolitan areas and 0.36 across states within a window. The labor-demand shock S is the prediction standardized across

[23] See Online Appendix Table OA.22.

[24] See Online Appendix Table OA.11, Part 2, columns 1 and 6.

markets within each window. It says how much more labor demand a market's industry mix predicts than the other markets' over the same ten years.[25]

Two components of demand can be measured without the price data and held fixed, the shock magnitude and the natural-amenity scale of McGranahan (1999), which takes the steep land share's place because it shares that share's topographic component. Holding either fixed leaves the state sending side in place, at 7.2 (2.3) and 7.9 (2.3) log points against 7.5. Amenity-rich states account for more of other states' variance in the raw network, 19.5 log points (6.0) per standard deviation of the scale, but not once the national factor is removed, 2.9 (4.8), as size does.[26] The timing of demand gives the sharper test, and it is a receiving-side test. Neither demand measure covers demand from consumption amenities or from local shocks unrelated to industry mix, so the sending-side result remains consistent with a supply reading without isolating it. I therefore read WRLURI as an observed characteristic of a market's land-use regime, with demand correlates and a state-institutional component, and not as a pure measure of supply constraints.

5.2 The receiving side, the national cycle and labor demand

In the raw network regulated states look like receivers of dependence, partly because they follow the national cycle more closely. A standard deviation of regulation is associated with 2.8 log points more external connectedness (Table 2, column 2), or 2.1 percentage points. The first principal component accounts for 8.1 percentage points more of a state's growth variance per standard deviation of the index, with a cluster-robust standard error of 1.2, and for 56 percent of that variance on average in the most regulated quarter of states against 35 percent in the least.[27] That is a fact about the price data and needs no network. Where housing supply is inelastic, national conditions show up in prices, and where it is elastic they show up in building, so the prices of supply-constrained markets should track the common cycle more closely (Glaeser et al., 2008; Guren et al., 2021; Tidwell et al., 2023). In a row-normalized decomposition a market that tracks the common factor more closely has more of its own row attributed to other markets, whatever its links to particular markets. The raw receiving side of regulated states is therefore partly the national cycle.

Whether regulated states also take more of their variance from other states beyond the cycle depends on how the network treats each state's own lags, and I do not settle it. In the benchmark network the factor-adjusted gradient is 1.7 log points (column 5), with an interval of $[-0.6, 4.0]$ that includes zero. The two networks that shrink the own lags less, with the own lags exempt from the penalty and with eight own lags, give 5.0 and 3.3, both distinguishable from zero, and the raw gradient is steeper there as well, 5.5 log points against 2.8. In none of the three networks is the

[25] Online Appendix Section OA.5.2 gives the construction, including the shock magnitude used as a control, the absolute deviation of predicted growth from the year's mean scaled within each industry-classification era.

[26] See Online Appendix Table OA.11, and Table OA.7 for two amplitude controls built from the price series themselves.

[27] See Online Appendix Table OA.3. A disturbance that spreads from one market to others within a month also loads on the first component, so the heavier loading of regulated states could partly be propagation along migration links.

change the purge makes to the receiving-side gradient distinguishable from zero.[28] The size of a receiving-side effect is therefore tied to a modeling choice about persistence, and three things hold in every network: regulated states load more heavily on the national factor, that loading raises their raw receiving side, and their sending side survives the purge.

The metropolitan system reads the same way, and it can hold a state's institutions fixed. Across the 100 largest metropolitan areas over the same years, a standard deviation of regulation is associated with 0.9 log points more external connectedness with Census-division effects and population controls, and, among the 64 areas of the twelve principal states with at least three sample areas, with 0.5 log points more under principal-state-by-year effects that absorb every shock common to a state's areas in a given year, with a wild cluster bootstrap p-value over the twelve states of 0.009. After the purge the two estimates are 0.7 and 0.3, and the within-state p-value is 0.33.[29] Differences in regulation among areas of one state hold its statutes, courts and lenders fixed. They go with how much of an area's variance comes from elsewhere in the raw network, and the association is imprecise beyond the national cycle. The steep land share, the second supply proxy, is imprecisely associated with every directional outcome at both scales and in both networks. Terrain does not do what regulation does.

Where a market's own demand is rising, regulated metropolitan areas take more of their forecast-error variance from other areas. That is the ordering a binding supply constraint implies. In the model a target with less elastic housing supply places more weight on other markets' prices, and with durable housing that difference appears only when the target's own demand rises. Larger demand shocks in regulated markets predict a different pattern: they raise the share of a market's forecast-error variance that is its own and so lower its external connectedness, whatever their sign. Table 3 puts the question to the labor-demand shock across the 100 metropolitan areas, on all three sides of an area's position, in the raw network and beyond the national cycle, in the layout of Table 2. Every column holds fixed the number of an area's same-state sample partners, which is strongly associated with external connectedness, and the climate part of the natural-amenity scale, since regulated markets are also the high-amenity places of the demand alternative.[30]

The receiving side is the test, and beyond the national cycle it shows the ordering a binding constraint implies (column 5). At the average shock the regulation gradient of log FROM is 0.53 log points per standard deviation (0.38), and it rises with the shock, by 0.78 log points per standard deviation of each (0.22). The magnitudes are small because external connectedness varies little

[28] From one regression that stacks each network's raw and factor-adjusted forms, the change is −1.1 (0.9), −0.6 (1.1) and, in the years in which both forms of the eight-own-lag network are stable, −0.2 (0.9). The levels are in Table 5, Panel A.

[29] See Online Appendix Table OA.12. Online Appendix Section OA.6 splits each area's external connectedness by the source's state: regulated areas take less of it from areas of their own state and more from areas in other states, in both networks, and part of that split rests on the areas that straddle a state line.

[30] The climate index sums the scale's January temperature, January sun, July temperature and July humidity components, averaged over an area's counties with 1990 population weights. Its topography component is left out because terrain enters on its own as the steep land share, and its water-area component with it. With the full scale and its interaction with the shock added to the climate control, the receiving-side interaction is 0.85 (0.24) in the specification without the steep land share terms.

Table 3: Regulation, relative labor demand and directional connectedness across metropolitan areas, raw and factor-adjusted networks, 1985–2025

	Raw			Factor-adjusted		
Dependent variable	log TO	log FROM	log NET	log TO	log FROM	log NET
	(1)	(2)	(3)	(4)	(5)	(6)
WRLURI 2006	6.15***	0.95***	5.20**	3.11	0.53	2.58
	(2.01)	(0.25)	(1.91)	(1.77)	(0.38)	(1.53)
Labor-demand shock S	−3.85**	−0.05	−3.81**	−2.39**	−0.18	−2.21**
	(1.44)	(0.17)	(1.44)	(1.06)	(0.20)	(0.99)
WRLURI × S	−1.00	0.01	−1.01	1.65	0.78***	0.87
	(1.25)	(0.17)	(1.22)	(1.00)	(0.22)	(0.93)
Steep land share	2.60	0.30	2.30	1.14	0.45	0.69
	(2.29)	(0.31)	(2.12)	(1.95)	(0.39)	(1.70)
Steep land share × S	0.46	0.02	0.44	0.58	−0.04	0.62
	(0.98)	(0.12)	(0.98)	(1.07)	(0.18)	(0.96)
Climate amenity index	−4.42	−2.06***	−2.35	−2.10	−2.14**	0.04
	(3.37)	(0.45)	(3.32)	(3.19)	(0.71)	(2.77)
Lagged log population	5.31**	0.29	5.02**	1.01	0.01	1.01
	(2.16)	(0.23)	(2.16)	(1.76)	(0.35)	(1.54)
Outside-node-set share	−5.27	−1.00***	−4.27	−1.93	0.37	−2.29
	(2.90)	(0.31)	(2.80)	(2.27)	(0.42)	(2.04)
Same-state sample partners	6.66***	0.96***	5.71***	8.79***	2.08***	6.71***
	(0.87)	(0.11)	(0.85)	(0.90)	(0.22)	(0.76)
Year and Census-division effects	✓	✓	✓	✓	✓	✓
Area-years	4,100	4,100	4,100	4,100	4,100	4,100
Within R^2	0.149	0.157	0.117	0.288	0.260	0.217

Note: The 100 largest metropolitan areas, 41 annual networks 1985–2025; raw network in columns (1)–(3), factor-adjusted in (4)–(6). $\log \text{NET} = \log(\text{TO}/\text{FROM})$, so its coefficients equal the TO minus the FROM coefficients. The labor-demand shock S is predicted labor demand, the shift-share prediction of an area's earnings growth over the window (Online Appendix Section OA.5.2), standardized across areas in each annual network. WRLURI 2006 and the steep land share are standardized across areas. The climate amenity index sums the four climate components of the USDA natural amenity scale; the outside-node-set share is the standardized share of the population within 200 km that lives in counties belonging to no sample area; same-state sample partners counts the other sample areas with the same principal state. Population is the CBSA's county population in the year before the network's year. Entries are 100 times the coefficients, approximately the percent difference in the outcome per standard deviation of a standardized regressor and 100 times the elasticity on a logged control. Within R^2 is net of the fixed effects. Cluster-robust standard errors by metropolitan area in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

across areas. Within a year the standard deviation of log FROM is 8.7 log points, so a standard deviation of the shock moves the regulation gradient by about a tenth of that spread. Larger own shocks would lower external connectedness whatever their sign, so the slopes above and below the year's average shock would sum to zero. They sum to 1.42 (0.43), with a slope of 1.24 (0.37) above the average and 0.19 (0.41) below it; their difference, 1.05 (0.65), is not distinguishable from zero, so the data do not separate a kink at the average from a gradient that rises with relative demand.[31] With area and division-by-year effects, which compare an area with itself relative to its division

[31] See Online Appendix Table OA.9, which reports the receiving side in four specifications, including the split and the specification without the steep land share terms, in which the interaction is 0.76 (0.21) and on which the checks of Section 6 are run. Split at the average, the steep land share's slopes have opposite signs, the pattern larger own shocks in either direction would produce.

in the same year, the interaction is 0.77 (0.32), with a p-value of 0.043, so it is not a difference between persistent types of place. Across states it is positive and imprecise, 1.4 (1.1). It is absent from the raw network, 0.01 (0.17) (column 2), where the receiving side mostly measures loading on the national factor, and the steep land share has no interaction beside regulation, −0.04 (0.18).

The other two sides show what that does to an area's place in the network. Beyond the national cycle, regulated areas with higher relative labor demand also account for somewhat more of other areas' variance, 1.65 (1.00) on log TO (column 4), so their net position barely moves, 0.87 (0.93) on log NET (column 6). The sending-side interaction is the less secure of the two. It is not distinguishable from zero, it falls to 0.8 (1.2) on an apportioned node set, and with area and division-by-year effects it is 1.15 (1.22) against the receiving side's 0.77.[32] The shock's own row compares types of place rather than demand conditions. At average regulation, areas with higher relative labor demand account for less of other areas' variance and sit further toward the receiving side, −2.39 (1.06) on log TO and −2.21 (0.99) on log NET, but with area and division-by-year effects the two are 0.19 (0.91) and −0.16 (0.89). The row describes which places have strong predicted demand, not what demand does to them. As on the sending side, the shock measures only the demand that an area's industry mix predicts. Other demand could produce the same ordering, so the receiving-side result is consistent with the supply multiplier but does not isolate it.

5.3 Migration and the allocation of dependence

Among metropolitan areas in different states, more of a market's dependence is attributed to the markets its households are linked to by migration. Across states, migration adds nothing to geography. Table 4 reports the two designs that decide this on the factor-adjusted networks. The first holds source-by-year and target-by-year effects fixed, and the second adds geography on top, which here includes the interactions of distance with contiguity and with region. Across states the association is present without geography and absent with it. A one-percentage-point increase in conditional migration exposure $x_{ij,t}$ is associated with 0.059 percentage points (0.017) more conditional connectedness beyond the national cycle under node-year effects alone (column 1), and with 0.000 (0.016) once geography enters (column 2). Conditional migration exposure has a standard deviation of 3.36 percentage points, so the slope without geography corresponds to 0.20 percentage points more conditional connectedness per standard deviation of exposure. The sequence behind the two columns starts from a pooled slope of 0.058, which geography alone lowers to 0.039 and geography with source-by-year effects to 0.006.[33]

Source-by-year effects absorb each source's general pull as a destination, which is largely its size, and leave the comparison among the sources of a given year. There a target does draw more dependence from the sources its migrants connect it to, but not more than distance, contiguity and region already predict. Moving costs shape both migration and proximity, and at state scale

[32] See Online Appendix Tables OA.8 and OA.10.

[33] Online Appendix Table OA.14 reports the sequence on the raw and the factor-adjusted networks. The raw network, in which the national cycle is part of every cell, starts from a stronger association and shows the same pattern.

exposure is close to a gravity function of the three, which alone account for 59 percent of the variance of conditional exposure within source-years and target-years.[34] Once geography enters, the association moves to distance. Among pairs that are neither contiguous nor in the same region, conditional connectedness falls by 0.16 percentage points (0.09) per log point of distance, and among contiguous pairs by a further 0.46 (0.23); the same-region interaction adds nothing. Target-by-year effects are a check at this scale, since the normalization fixes the row sums.

Table 4: Migration exposure and the allocation of connectedness across sources, factor-adjusted networks, 2005–2025

Dependent variable	Conditional connectedness share (percentage points)			
	States		Metropolitan areas	
	(1)	(2)	(3)	(4)
Exposure, pairs in different states	0.059^{***}	0.000	0.084^{***}	0.067^{***}
	(0.017)	(0.016)	(0.020)	(0.022)
Exposure, same-state pairs			−0.003	−0.024
			(0.026)	(0.032)
Log distance		-0.159^{*}		0.009
		(0.089)		(0.036)
Log distance × contiguity		-0.456^{**}		-0.252^{***}
		(0.226)		(0.072)
Log distance × same region		0.059		−0.052
		(0.232)		(0.049)
Log distance, same-state pairs				-0.625^{*}
				(0.329)
Other geography controls	–	✓	–	✓
Source-by-year effects	✓	✓	✓	✓
Target-by-year effects	✓	✓	✓	✓
Target-by-year × same-state effects	–	–	✓	✓
Directed pair-years	47,376	47,376	207,522	207,522

Note: The 21 annual factor-adjusted networks 2005–2025; 48 target states with 47 sources each and 100 target metropolitan areas with 99 sources each, all pairs. The dependent variable is the conditional connectedness share, in percentage points of the target's external connectedness, and the regressor of interest is conditional migration exposure, the conditional migration share centered at $100/(N-1)$ within target and year, from IRS flows pooled over $t-15$ through $t-11$. Conditional shares average 2.13 percentage points across states and 1.01 across metropolitan pairs, 0.71 across pairs of areas in different states. Columns (1) and (3) have source-by-year and target-by-year effects, and columns (2) and (4) add log distance, contiguity, same Census region and their pairwise interactions on top; at the metropolitan scale contiguity is a shared Census division. The other geography controls, contiguity, same region and, across states, their interaction, are included and not shown, so the log-distance coefficient is the distance slope for pairs that are neither contiguous nor in the same region and each interaction is the additional slope for contiguous pairs or for pairs in the same region. The metropolitan columns keep every pair and give the pairs of areas with the same principal state, the state holding the largest share of an area's 2010 housing units, a target-by-year effect of their own, which absorbs whatever part of each area's external connectedness stays inside its state, and let the exposure and distance slopes differ for those pairs; the rows report the slopes for pairs of areas in different states and for same-state pairs, and every state pair is a pair in different states. The 378 pair-years of the 18 areas with a single same-state partner are fitted exactly by their own effect and drop out. Online Appendix Table OA.14 reports the sequence of designs behind these columns, on the raw and the factor-adjusted networks and on the out-of-state subsample, and Online Appendix Table OA.15 the metropolitan columns on both networks. Shared-node standard errors in parentheses, which allow two directed pairs to be dependent whenever they share a node; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 against Student's t with one degree of freedom fewer than the number of nodes.

[34] Online Appendix Table OA.18 reports the geography benchmarks with and without exposure and the fit that exposure adds in each design.

Among metropolitan areas the association survives node-year effects and geography, and it is read from pairs of areas in different states. The metropolitan columns repeat the regressions across all pairs of the 100 metropolitan areas, with exposure built from county-to-county flows under the same pool and with shared Census division replacing contiguity. Pairs of areas with the same principal state share state-level shocks, institutions and lenders as well as migrants, and the one-factor purge leaves state-level common structure in place; they are 3.7 percent of dyads and carry 19 percent of the variance of exposure. Rather than drop them, the regressions give the same-state sources of each target-year a level of their own, which absorbs whatever part of an area's external connectedness stays inside its state, and let the exposure and distance slopes differ for those pairs. For pairs in different states the slope is 0.084 (0.020) with node-year effects (column 3) and 0.067 (0.022) once geography is added (column 4). A pair whose exposure is one percentage point above an equal split therefore carries 0.07 points more of the target's external connectedness, about 9 percent of the 0.71-point average share of such pairs. Among same-state pairs there is no exposure gradient at all, −0.02 (0.03), so what a same-state source carries is the within-state level and not its migration share. Estimated on the pairs in different states alone, the slope is 0.060 (0.025), and in the sequence behind it 0.077 pooled, 0.069 with geography and 0.070 with source-year effects.[35] Distance orders shares within divisions. Among pairs in different regions conditional connectedness does not vary with distance, 0.01 (0.04), while within a division it falls by 0.24 percentage points per log point of distance, the interaction being −0.25 (0.07), and among same-state pairs by 0.63 (0.33).

What survives among pairs in different states is an association between exposure and the pairwise pattern of dependence across links, which distance, division and region capture only in part; industrial similarity, social ties and cross-border proximity like that of New York and Bridgeport could also produce it. Three further comparisons bound it. The association is carried by the direct part of exposure, the moves between the two markets, 0.064 (0.029) on pairs in different states under the geography and node-year design, while the part that runs through shared origins, which is what distinguishes the model's exposure from a simple flow measure, is 0.013 (0.050), too imprecise to include or exclude. Nearby pairs carry part of the precision: dropping the out-of-state pairs whose internal points lie within 100 miles of each other lowers the out-of-state estimate from 0.060 to 0.047 (0.025), with an interval of $[-0.002, 0.095]$, so most of the point estimate remains without them but it is no longer distinguishable from zero. And the designs that ask whether the association has a direction, unordered-pair effects alone and with node-year effects, find nothing distinguishable from zero at either scale, because exposure barely varies within a pair; the data cannot say that migration makes one market's dependence on another larger than the reverse, and they cannot say that it does not.[36]

[35] Online Appendix Table OA.14 reports the sequence on pairs in different states and on all pairs without the same-state terms; without them the all-pairs slope, 0.171 (0.056), carries the within-state level. Online Appendix Table OA.15 reports these columns on both networks. Giving every target the same same-state level instead returns a slope of 0.094 and a positive distance coefficient, because targets with one to eleven same-state partners do not share one level. In the raw network the slope for pairs in different states is 0.012 (0.007).

[36] See Online Appendix Section OA.7, Tables OA.20 and OA.17.

Regulation does not change which markets a market's dependence comes from, as the model predicts. The supply multiplier scales every weight of a target's row and cancels when the row is normalized, so the model implies no interaction of exposure with the target's regulation in conditional shares. No such interaction, nor one with the steep land share, is distinguishable from zero across states or among metropolitan areas in different states, on either node set. Among metropolitan pairs in different states the regulation interaction is 0.007 (0.015) against an exposure slope of about 0.06. The data rule out interactions as large as the slope but not ones half its size.[37]

6 Robustness

The sending side of regulated states is the most secure result of the paper and the receiving side beyond the national cycle the least, and the pattern of what survives says something about what each side is. Tables 5 and 6 report both sides under alternative networks, second-stage specifications, samples and timing. The checks on the allocation result are in the Online Appendix and are summarized where they bear on the text.

6.1 Alternative networks

The network enters every result in this paper, and how it treats each state's own history decides how much of a state's forecast error is attributed to others. The benchmark penalizes own lags exactly as it penalizes cross-state lags, and it leaves the fitted residuals serially correlated. The median absolute lag-one autocorrelation of the in-sample residuals is 0.39, against 0.86 for monthly price growth itself, whose repeat-sales indexes are smooth. I therefore re-estimate the directional results on two networks that differ from the benchmark in the own-lag treatment alone, each in its raw and its factor-adjusted form. The first exempts the two own lags from the penalty, at the same lag order, window, mixing parameter, validation design and path. The second also extends the own lags to eight, the shortest own-lag order at which residual serial dependence becomes weak.[38] The two alternatives are checks on the benchmark and not candidates for it.[39]

How much of other states' variance is attributed to a state does not depend on how much persistence the network allows each state (Table 5, Panel A). In the factor-adjusted networks the TO association is 7.5 log points under the benchmark, 9.1 with the own lags free and 7.1 with eight own lags in that network's stable years, and in the raw networks 10.5, 13.7 and 10.7. Log NET is

[37] See Online Appendix Table OA.16.

[38] Freeing the own lags lowers the in-sample median residual autocorrelation from 0.38 to 0.21 and eight own lags to 0.04, and the active cross-state slopes per equation fall from 25.4 to 13.2 and 6.7 in the annual networks from 2005. The eight-own-lag system is not stable in the raw annual networks of 1985 and 2002 through 2009, the boom and bust of the 2000s, so its columns are estimated on its stable years, and the table repeats the benchmark and own-lags-free regressions on those years beside it. The metropolitan results rest on the benchmark network alone, because the metropolitan system with its own lags exempt from the penalty is unstable in many windows. See Online Appendix Table OA.2 and Section OA.3.4.

[39] See Appendix B.1.

Table 5: Sending and receiving sides under alternative networks and second-stage specifications, raw and factor-adjusted

	log TO		log FROM		log NET	
	Raw (1)	Factor-adjusted (2)	Raw (3)	Factor-adjusted (4)	Raw (5)	Factor-adjusted (6)
Panel A. Alternative networks						
Benchmark network	10.54**	7.50***	2.76***	1.67	7.78**	5.83**
	(3.62)	(2.32)	(0.60)	(1.05)	(3.40)	(2.08)
Own-lags-free network	13.73***	9.14***	5.52***	4.95***	8.21**	4.18**
	(3.26)	(1.86)	(1.17)	(1.23)	(3.15)	(1.62)
Benchmark, eight-own-lag stable years	9.06**	8.10***	2.19***	1.61	6.87	6.49***
	(3.93)	(2.32)	(0.61)	(1.02)	(3.75)	(2.07)
Own-lags-free, eight-own-lag stable years	13.09***	9.56***	4.53***	5.05***	8.56**	4.51***
	(3.27)	(1.85)	(1.11)	(1.21)	(3.07)	(1.60)
Eight-own-lag network, its stable years	10.69**	7.15***	2.61**	3.26**	8.08**	3.88**
	(3.59)	(2.22)	(1.04)	(1.09)	(2.98)	(1.50)
Annual networks, rows 1–2	41	41	41	41	41	41
rows 3–5	32	38	32	38	32	38
Benchmark estimator, more common components removed						
Bai–Ng number of components removed	–	1.41	–	−0.50	–	1.91*
		(1.14)		(0.62)		(0.96)
Two components removed	–	5.36**	–	1.28	–	4.08**
		(2.06)		(0.79)		(1.71)
First component and its score at lags 1–3 removed	–	5.19**	–	1.08	–	4.11**
		(2.12)		(0.88)		(1.59)
First component and its score at lags 1–6 removed	–	8.58***	–	1.25	–	7.34***
		(2.20)		(0.85)		(1.76)
Panel B. Alternative second-stage specifications, benchmark network						
Census-region effects	21.06***	11.46***	2.17*	1.92	18.89***	9.55***
	(5.40)	(3.06)	(1.05)	(1.53)	(5.63)	(3.00)
WRLURI 2018 in place of 2006	7.44	6.30**	2.15**	0.90	5.30	5.40**
	(4.01)	(1.86)	(0.71)	(1.05)	(3.78)	(1.87)
Saiz unavailable-land share added	10.93**	7.70***	2.61***	1.71	8.33**	5.99**
	(3.66)	(2.40)	(0.67)	(1.15)	(3.37)	(2.05)

Note: 48 states. Each cell is 100 times the WRLURI 2006 coefficient in the regressions of Table 2 with the change the row names; odd columns raw, even columns factor-adjusted. Panel A varies the network. Rows 1–2 use the 41 annual networks 1985–2025 and rows 3–5 the years in which the eight-own-lag network is dynamically stable, 32 raw and 38 factor-adjusted. The last four rows remove, in each window and inside each validation fold, the number of principal components the Bai and Ng (2002) criterion selects on the window's raw growth (a median of four), two components, or the first component together with its score at the stated lags; they have no raw form. Panel B adds Census-region effects, replaces the 2006 index with the 2018 one (each standardized by its own cross-state standard deviation) or adds the Saiz unavailable-land share. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

5.8, 4.2 and 3.9 in the factor-adjusted networks, each distinguishable from zero at the 5-percent level. How much a state receives does depend on it, as Section 5.2 reports.

Removing more common structure than one factor weakens the sending side, and what it leaves says what the sending side is. The one-factor purge leaves regional common structure in place, and Census-region effects absorb the regional level of TO but not within-region differences in how strongly states load on a regional component.[40] Removing the first component together with its score at lags one to three or one to six leaves the factor-adjusted TO gradient at 5.2 (2.1) and 8.6 (2.2), and removing two components at every window leaves 5.4 (2.1), with an interval of $[0.9, 9.9]$. Removing the number of components the Bai and Ng (2002) criterion selects on the window's raw growth, a median of four between two and eight, leaves 1.4 (1.1), with an interval of $[-1.3, 3.7]$, and a log NET gradient of 1.9 (1.0).[41] Two readings fit. Either the factor-adjusted sending side reflects regulated states' position within regional clusters, whose shared shocks are attributed to each member, and not structure among markets, or the larger purge removes real propagation that loads on several factors, which makes it a demanding test. The design cannot choose between them.

6.2 Second-stage specifications and inference

Regulated states are not spread evenly across regions, so the association could be a regional difference. It is not. Adding Census-region effects (Panel B of Table 5) raises the log TO and log NET estimates, to 21.1 and 18.9 in the raw network and 11.5 and 9.5 in the factor-adjusted one, and moves the external-connectedness estimate little, to 2.17 raw and 1.92 factor-adjusted. The sending-side association therefore appears among states within the same broad region, where it is if anything stronger. With the 2018 index in place of the 2006 one, each vintage standardized by its own cross-state standard deviation, the factor-adjusted TO association is 6.3 log points and log NET 5.4, both distinguishable from zero, while the raw TO association, 7.4, and raw log NET, 5.3, are not. The 2018 survey postdates every window ending before 2018 and measures the regime the early windows faced with more error, and the raw coefficients lose their precision first. Adding Saiz's unavailable-land share leaves the factor-adjusted TO association at 7.7. How the index is aggregated or conditioned on population changes little, the factor-adjusted TO association staying between 7.5 and 8.0 log points, and terrain stays unassociated with either side of a market's position however it is measured.[42]

Neighboring states share shocks, and allowing for that does not overturn the sending-side gradient. Clustered by the nine Census divisions, the wild bootstrap interval for the factor-adjusted TO coefficient widens to $[1.1, 16.4]$, from $[2.7, 12.7]$ by state, and for log NET it is $[3.0, 11.9]$. Conley standard errors are smaller than the state-clustered ones.[43] Metropolitan areas that share a dominant industry are exposed to the same demand shocks, so their errors may be correlated,

[40] See Appendix C.

[41] See Online Appendix Table OA.6.

[42] See Online Appendix Tables OA.21 and OA.5, and Online Appendix Section OA.1.3.

[43] The Conley standard errors allow correlation within 500, 1,000 or 1,500 kilometers, and within a state at any lag. See Online Appendix Table OA.23.

and the metropolitan receiving-side interaction survives allowing for that. Clustered by each area's dominant industry in 1975, 16 clusters, its wild bootstrap p-value is 0.010, and that of the regulation slope above the average is 0.037.[44]

6.3 Alternative samples

Table 6: Sending and receiving sides under alternative samples and timing, raw and factor-adjusted

	log TO		log FROM		log NET	
	Raw (1)	Factor-adjusted (2)	Raw (3)	Factor-adjusted (4)	Raw (5)	Factor-adjusted (6)
Panel A. Alternative samples, benchmark network						
Without the Northeast, 39 states	16.00**	11.15**	3.22**	2.72	12.78**	8.43**
	(5.26)	(4.22)	(1.03)	(1.93)	(4.92)	(3.17)
Without the Midwest, 36 states	10.01**	7.00**	2.67***	1.71	7.34*	5.29**
	(3.88)	(2.60)	(0.78)	(1.29)	(3.68)	(2.35)
Without the South, 32 states	8.59	6.71**	1.91**	0.36	6.68	6.34**
	(5.09)	(2.72)	(0.65)	(0.99)	(4.96)	(2.73)
Without the West, 37 states	13.79***	10.42***	3.04***	1.94	10.75**	8.47***
	(3.90)	(2.40)	(0.78)	(1.41)	(3.69)	(2.07)
Each state omitted in turn, range	[9.05, 11.83]	[6.38, 8.57]	[2.52, 3.09]	[0.92, 2.01]	[6.50, 9.07]	[4.99, 7.18]
Panel B. Alternative timing, benchmark network						
Annual networks 1985–2004	10.50**	7.43***	2.50**	0.56	8.00**	6.87***
	(3.32)	(2.12)	(1.07)	(1.13)	(3.24)	(1.89)
Annual networks 2005–2025	10.73*	7.67**	3.06***	2.75*	7.66	4.92
	(5.92)	(3.43)	(0.58)	(1.37)	(5.94)	(3.38)
Annual networks 2010–2025	6.84	5.38	2.23***	1.15	4.61	4.23
	(6.73)	(3.33)	(0.58)	(1.44)	(6.82)	(3.39)

Note: 48 states. Each cell is 100 times the WRLURI 2006 coefficient in the regressions of Table 2 with the change the row names; odd columns raw, even columns factor-adjusted. Panel A removes the states of one Census region at a time, on the national networks, which are not re-estimated, with covariates standardized across all 48 states; its last row gives the smallest and largest estimate when each state is removed in turn. Panel B restricts the annual networks to the named years. Cluster-robust standard errors by state in parentheses; *, ** and *** denote $p < 0.10$, 0.05 and 0.01 from a wild cluster bootstrap-t that imposes the null (9,999 draws).

No single state or region produces the sending-side association. Omitting each of the 48 states in turn leaves the factor-adjusted TO estimate between 6.4 and 8.6 log points, and omitting each Census region in turn between 6.7 (without the South) and 11.1 (without the Northeast). Each of the four region omissions leaves a factor-adjusted TO and log NET association distinguishable from zero at the 5-percent level, whereas in the raw network the TO association without the South, 8.6 (5.1), is not (Table 6, Panel A). The receiving side is more fragile on this margin, as it is on the estimator, and it leans on the South. Omitting the South lowers the raw external-connectedness estimate from 2.76 to 1.91 log points and the factor-adjusted one from 1.67 to 0.36. These are influence calculations on fixed national networks, which are not re-estimated after a state or region is removed.[45]

[44]The checks use the specification without the steep land share terms and its split at the year's average shock, Online Appendix Table OA.9, columns (1) and (4). With area clusters the two p-values are 0.001 and 0.023. See Online Appendix Table OA.8.

[45]See Online Appendix Table OA.23.

The receiving-side interaction is not produced by a few high-demand places. It stays between 0.63, without the Mountain division, and 0.94, without the West North Central, when any single Census division is left out, and it is 0.68 (0.22) when San Jose, San Francisco, Seattle, Boston and Austin are left out together. The metropolitan node set is itself a sample choice, and it bears on the economics. Ranking areas by 2000 population puts eleven of the 100 in California, eight in Florida, seven in Ohio and six each in New York, Pennsylvania and Texas, and leaves nine states without an area, while areas in the same state share state shocks that the one-factor purge leaves in place. On an apportioned node set of 100 areas that gives each of the 48 contiguous states at least one area and no state more than six, the allocation result holds, the receiving-side interaction keeps its sign and most of its size but not its precision, 0.60 (0.34) against 0.76 (0.21), and the sending-side interaction does not survive, 0.8 (1.2) against 1.9 (0.9), so that interaction is specific to the benchmark node set and I do not lean on it.[46]

6.4 Alternative timing

Estimated as a separate 48-state cross-section in every annual network, the factor-adjusted TO coefficient is positive in 32 of the 41 years, against 33 raw, and its interval lies above zero in 10, against 12 raw. In no year does the interval lie below zero, in either network or for either sending-side outcome. Averaged by decade, the factor-adjusted coefficient is 7.3 log points over 1985–1989, 9.2 in the 1990s, 9.4 in the 2000s, 6.9 in the 2010s and 2.9 since 2020, and the raw one is about 15 in the 1990s and 2000s and 0.9 since 2020.[47] The sending-side association is therefore present from the first windows, largest in the raw network in the windows that contain the boom and bust of the 2000s, and weakest in both networks in the most recent windows. That is consistent with a national cycle that has come to dominate. The common component accounts for 69 percent of a state's growth variance in the windows since 2020, against 22 percent in those ending in 1985–1989, and a smaller residual variance leaves less room for differences between states.[48] Split at 2005 (Panel B of Table 6), the factor-adjusted TO association is 7.4 before and 7.7 after, and from 2010 it is 5.4 (3.3). The factor-adjusted net position is 6.9 before 2005, but 4.9 from 2005 and 4.2 from 2010, neither distinguishable from zero. The sending-side association is therefore present in both halves of the period, and the net position mainly in the first. The receiving-side interaction holds across both industry classifications, 0.74 (0.23) in the NAICS period and 0.96 (0.27) in the SIC period, and holding the migration network fixed at its 1990–1994 pool leaves the allocation results in place.[49]

The timing of the index raises a last question, whether regulation responds to the price histories it is compared with. WRLURI 2006 postdates the windows ending in 1985–2005, so regulation could partly respond to past price growth if places that saw fast or volatile growth later adopted tighter rules, and no state measure of regulation before 1990 is available to check this. Land-use

[46] Both comparisons use the specification without the steep land share terms. See Online Appendix Table OA.10, which reports the comparison on apportioned node sets of 100 and 75 areas.

[47] See Online Appendix Figure OA.3 and Section OA.4.2, which also reports the receiving side year by year.

[48] See Appendix Figure C.1.

[49] See Online Appendix Tables OA.8 and OA.19.

restrictions accumulate over decades, so the 2006 index records a slow-moving regime rather than a recent policy, and the labor-demand literature the paper follows treats it as a determinant of local housing-supply elasticity (Diamond, 2016). The regressions compare states and not dated treatments, and they claim association, not causation, so the supply reading is an interpretation of the association and not something the design establishes. Supply elasticities also changed over the period, falling between the booms of the 2000s and the 2010s where regulation tightened (Aastveit et al., 2023), which is one reason the 2018 index is carried as a check.

7 Conclusion

What connects local housing markets? Measured as a directed network of forecast-error dependence, they are strongly and increasingly connected. In the raw state network about two-thirds of a market's forecast-error variance was attributed to other markets in the late 1980s and more than 90 percent since 2020. Some markets account for several times as much of other markets' variance as others do, and dependence falls with distance, by less over time. Most of this is a common component whose weight has grown since the 2000s. It accounts for almost all of the rise in measured connectedness, for the flattening of the distance gradient and for much of the spread in net positions. The purge cannot say whether its growth reflects common shocks or propagation among markets. Dependence orthogonal to the component has not grown, and it is not random.

More of other states' forecast uncertainty beyond the national cycle is traced to tightly regulated states, 7.5 log points more per standard deviation of the index in the benchmark network, and regulated states sit closer to being net sources. The association keeps its sign under every alternative network, sample and timing split, stays precise under each own-lag treatment, within Census regions and without any single state or region, and its limits say what it is. It is weakest in the windows since 2020, when the national cycle leaves least room for differences between states; it is imprecise on average across metropolitan areas beyond the cycle; and it does not survive removing the larger number of common factors an information criterion selects in each window, so the factor-adjusted sending side may partly reflect regulated states' position within regional clusters. The receiving side tells a different story. Regulated states track the national cycle more closely, which raises their receiving side in the raw network, and whether they receive more beyond the cycle depends on how the autoregression shrinks each state's own lags, which the paper leaves open. Beyond the cycle, regulated metropolitan areas take more of their forecast-error variance from other areas where labor demand predicted from industry mix is strong, the ordering a binding supply constraint implies once durable housing is added to the model; the interaction holds under industry clustering, division and area omissions and both industry classifications, and keeps its sign, with less precision, on an apportioned node set. Steep terrain shows no such interaction. Across states, migration exposure adds nothing to distance, contiguity and region once source-by-year effects are held fixed, because at that scale it is close to a gravity function of the three. Across metropolitan areas in different states it lines up with which markets a market's dependence comes from, the association survives

node-year effects in either node set, and it is carried by direct migration links, loses precision when nearby pairs are dropped and has no direction the data can detect within pairs.

These results give an economic reading to a reduced-form housing network. Of the framework's two comparative statics, the allocation prediction holds among metropolitan areas in different states, where migration orders a market's sources and regulation leaves that ordering alone, and the external-weight prediction is found beyond the national cycle only across states and only in the networks that shrink own lags less, while the receiving side follows the timing the model implies once durable housing is added. The sending-side regularity, the most secure result of the paper, lies outside the formal model. I read it as a supply constraint that turns a market's own demand shocks into price movements that other markets respond to, and the evidence supports that reading without isolating it from larger demand shocks in regulated markets. The variance shares measure forecast-error dependence rather than structural transmission, the regulation index describes a market's land-use regime together with its demand and state-institutional content, and every interval conditions on the estimated network. Two extensions would sharpen the results. A national-plus-regional de-factoring along the lines of Bailey et al. (2016) would say how much of the sending-side pattern is structure among markets rather than regional common shocks, and carrying first-stage estimation uncertainty into the second stage would make the conditional intervals reported here unconditional.

Data availability

The house-price and migration data are publicly available from Freddie Mac (Freddie Mac, 2026a) and the IRS Statistics of Income (Internal Revenue Service, 2024). The land-unavailability and Wharton regulation measures are available from their original authors (Lutz and Sand, 2026; Gyourko et al., 2008, 2021). State and county population estimates are public releases of the Bureau of Economic Analysis and the Census Bureau Population Estimates Program. The industry wages and earnings behind the labor-demand proxies are BEA regional accounts (U.S. Bureau of Economic Analysis, 2025), the natural-amenity scale is a USDA release (U.S. Department of Agriculture, Economic Research Service, 1999), and metropolitan county composition follows the 2023 Census delineation (U.S. Census Bureau, 2023). The replication package contains the code that estimates the networks and the second-stage regressions, the estimated networks, and the programs that build every table and figure of this paper from them. It also contains the programs that estimate the monthly networks, with their inputs, and the inputs to the comparisons between the state and metropolitan systems. The state steep land share can be rebuilt from the CBSA steep land shares and the Census population and housing-stock counts in the package, including the Colorado grouping and the Montana allocation. Some inputs to the metropolitan comparisons are built by separate programs, documented with their analysis panels. The package also contains the bootstrap draws, input checksums and execution logs of the state-cluster and network-sensitivity runs. Rebuilding the exhibits from the stored networks is separate from re-estimating every network from the raw data.

References

Aastveit, K. A., Albuquerque, B., and Anundsen, A. K. (2023). Changing supply elasticities and regional housing booms. *Journal of Money, Credit and Banking*, 55(7):1749–1783.

Antonakakis, N., Chatziantoniou, I., Floros, C., and Gabauer, D. (2018). The dynamic connectedness of UK regional property returns. *Urban Studies*, 55:3110–3134.

Antonakakis, N., Chatziantoniou, I., and Gabauer, D. (2021). A regional decomposition of US housing prices and volume: market dynamics and portfolio diversification. *The Annals of Regional Science*, 66:279–307.

Bai, J. and Ng, S. (2002). Determining the number of factors in approximate factor models. *Econometrica*, 70(1):191–221.

Bailey, M., Cao, R., Kuchler, T., and Stroebel, J. (2018). The economic effects of social networks: Evidence from the housing market. *Journal of Political Economy*, 126(6):2224–2276.

Bailey, N., Holly, S., and Pesaran, M. H. (2016). A two-stage approach to spatio-temporal analysis with strong and weak cross-sectional dependence. *Journal of Applied Econometrics*, 31(1):249–280.

Barigozzi, M., Cho, H., and Owens, D. (2024). FNETS: Factor-adjusted network estimation and forecasting for high-dimensional time series. *Journal of Business & Economic Statistics*, 42(3):890–902.

Barigozzi, M. and Hallin, M. (2017). A network analysis of the volatility of high dimensional financial series. *Journal of the Royal Statistical Society Series C: Applied Statistics*, 66(3):581–605.

Bartik, T. J. (1991). *Who Benefits from State and Local Economic Development Policies?* W. E. Upjohn Institute for Employment Research, Kalamazoo, MI.

Bergmeir, C., Hyndman, R. J., and Koo, B. (2018). A note on the validity of cross-validation for evaluating autoregressive time series prediction. *Computational Statistics & Data Analysis*, 120:70–83.

Brady, R. R. (2014). The spatial diffusion of regional housing prices across U.S. states. *Regional Science and Urban Economics*, 46:150–166.

Buchwalter, B., Diebold, F. X., and Yilmaz, K. (2026). Clustered network connectedness: A new measurement framework with application to global equity markets. *Journal of Econometrics*, page 106243.

Burman, P., Chow, E., and Nolan, D. (1994). A cross-validatory method for dependent data. *Biometrika*, 81(2):351–358.

Cameron, A. C., Gelbach, J. B., and Miller, D. L. (2008). Bootstrap-based improvements for inference with clustered errors. *Review of Economics and Statistics*, 90(3):414–427.

Case, K. E. and Shiller, R. J. (1987). Prices of single-family homes since 1970: New indexes for four cities. *New England Economic Review*, pages 45–56. September/October issue. Also NBER Working Paper 2393, doi:10.3386/w2393.

Chiang, M.-C. and Tsai, I.-C. (2016). Ripple effect and contagious effect in the US regional housing markets. *The Annals of Regional Science*, 56(1):55–82.

Cohen, J., Coughlin, C., and Soques, D. (2024). Housing price cycle interdependencies and comovement: A Markov-switching approach. *Journal of Real Estate Research*, 46(2):159–188.

Cohen, J. P., Ioannides, Y. M., and Thanapisitikul, W. (2016). Spatial effects and house price dynamics in the USA. *Journal of Housing Economics*, 31:1–13.

Cotter, J., Gabriel, S., and Roll, R. (2015). Can housing risk be diversified? a cautionary tale from the housing boom and bust. *Review of Financial Studies*, 28(3):913–936.

Cun, W. and Pesaran, M. H. (2022). A spatiotemporal equilibrium model of migration and housing interlinkages. *Journal of Housing Economics*, 57:101839.

Davidoff, T. (2016). Supply constraints are not valid instrumental variables for home prices because they are correlated with many demand factors. *Critical Finance Review*, 5(2):177–206.

Del Negro, M. and Otrok, C. (2007). 99 luftballons: Monetary policy and the house price boom across U.S. states. *Journal of Monetary Economics*, 54:1962–1985.

Demirer, M., Diebold, F. X., Liu, L., and Yilmaz, K. (2018). Estimating global bank network connectedness. *Journal of Applied Econometrics*, 33:1–15.

Di Maggio, M. and Kermani, A. (2017). Credit-induced boom and bust. *Review of Financial Studies*, 30(11):3711–3758.

Diamond, R. (2016). The determinants and welfare implications of US workers' diverging location choices by skill: 1980-2000. *American Economic Review*, 106:479–524.

Diebold, F. X. and Yilmaz, K. (2009). Measuring financial asset return and volatility spillovers, with application to global equity markets. *The Economic Journal*, 119:158–171.

Diebold, F. X. and Yilmaz, K. (2012). Better to give than to receive: Predictive directional measurement of volatility spillovers. *International Journal of Forecasting*, 28:57–66.

Diebold, F. X. and Yilmaz, K. (2014). On the network topology of variance decompositions: Measuring the connectedness of financial firms. *Journal of Econometrics*, 182:119–134.

Fairchild, J., Ma, J., and Wu, S. (2015). Understanding housing market volatility. *Journal of Money, Credit and Banking*, 47(7):1309–1337.

Ferreira, F. and Gyourko, J. (2023). Anatomy of the beginning of the housing boom across U.S. metropolitan areas. *Review of Economics and Statistics*, 105(6):1442–1447. Working paper version NBER Working Paper 17374, 2011.

Findley, D. F., Monsell, B. C., Bell, W. R., Otto, M. C., and Chen, B.-C. (1998). New capabilities and methods of the X-12-ARIMA seasonal adjustment program. *Journal of Business & Economic Statistics*, 16(2):127–152.

Foster, D. D. and Summers, A. A. (2005). Current state legislative and judicial profiles on land-use regulations in the U.S. Working Paper 512, Wharton Real Estate Center, University of Pennsylvania.

Freddie Mac (2026a). Freddie Mac house price index. Public data series.

Freddie Mac (2026b). Primary mortgage market survey: 30-year fixed rate mortgage average in the United States. FRED series MORTGAGE30US, Federal Reserve Bank of St. Louis.

Gabauer, D., Gupta, R., Marfatia, H. A., and Miller, S. M. (2024). Estimating U.S. housing price network connectedness: Evidence from dynamic elastic net, lasso, and ridge vector autoregressive models. *International Review of Economics & Finance*, 89:349–362.

Glaeser, E. L. and Gyourko, J. (2005). Urban decline and durable housing. *Journal of Political Economy*, 113:345–375.

Glaeser, E. L., Gyourko, J., and Saiz, A. (2008). Housing supply and housing bubbles. *Journal of Urban Economics*, 64:198–217.

Goetzmann, W. N. (1992). The accuracy of real estate indices: Repeat sale estimators. *Journal of Real Estate Finance and Economics*, 5(1):5–53.

Guren, A. M., McKay, A., Nakamura, E., and Steinsson, J. (2021). Housing wealth effects: The long view. *The Review of Economic Studies*, 88(2):669–707.

Gyourko, J., Hartley, J. S., and Krimmel, J. (2021). The local residential land use regulatory environment across U.S. housing markets: Evidence from a new Wharton index. *Journal of Urban Economics*, 124:103337.

Gyourko, J., Saiz, A., and Summers, A. (2008). A new measure of the local regulatory environment for housing markets: The Wharton residential land use regulatory index. *Urban Studies*, 45(3):693–729.

Hernández-Murillo, R., Owyang, M. T., and Rubio, M. (2017). Clustered housing cycles. *Regional Science and Urban Economics*, 66:185–197.

Holly, S., Pesaran, M. H., and Yamagata, T. (2010). A spatio-temporal model of house prices in the USA. *Journal of Econometrics*, 158(1):160–173.

Huang, H. and Tang, Y. (2012). Residential land use regulation and the US housing price cycle between 2000 and 2009. *Journal of Urban Economics*, 71(1):93–99.

Huang, Y.-L., Katselas, D., and Lin, P.-T. (2024). Unpacking housing market connectedness. Technical report, European Real Estate Society (ERES).

Internal Revenue Service (2024). Statistics of income tax stats: Migration data. Annual state-to-state migration files. Returns, exemptions, and adjusted gross income flows used for 1990–2022.

Kallberg, J. G., Liu, C. H., and Pasquariello, P. (2014). On the price comovement of U.S. residential real estate markets. *Real Estate Economics*, 42(1):71–108.

Karamon, K. (2023). Seasonal adjustment in house price indices in the COVID-19 period: A study of residual seasonality. Research paper, Freddie Mac.

Kim, Y. S. and Rous, J. J. (2012). House price convergence: Evidence from US state and metropolitan area panels. *Journal of Housing Economics*, 21(2):169–186.

Koop, G., Pesaran, M. H., and Potter, S. M. (1996). Impulse response analysis in nonlinear multivariate models. *Journal of Econometrics*, 74:119–147.

Landier, A., Sraer, D., and Thesmar, D. (2017). Banking integration and house price co-movement. *Journal of Financial Economics*, 125(1):1–25.

Lee, W. S. and Ma, E. (2025). Housing market connectedness and transmission of monetary policy. *Economic Inquiry*, 63:961–984.

Louie, S., Mondragon, J., and Wieland, J. (2025). Supply constraints do not explain house price and quantity growth across U.S. cities. Working Paper 2025-06, Federal Reserve Bank of San Francisco.

Lütkepohl, H. (2005). *New Introduction to Multiple Time Series Analysis*. Springer, Berlin.

Lutz, C. and Sand, B. (2026). Highly disaggregated land unavailability. Technical Report 4543310, SSRN.

MacKinnon, J. G. and Webb, M. D. (2017). Wild bootstrap inference for wildly different cluster sizes. *Journal of Applied Econometrics*, 32(2):233–254.

Malone, T. (2017). Housing market spillovers in a system of cities. Working paper, SSRN 3045823.

McGranahan, D. A. (1999). Natural amenities drive rural population change. Agricultural Economic Report 781, U.S. Department of Agriculture, Economic Research Service.

Mian, A. and Sufi, A. (2009). The consequences of mortgage credit expansion: Evidence from the U.S. mortgage default crisis. *Quarterly Journal of Economics*, 124(4):1449–1496.

Miao, H., Ramchander, S., and Simpson, M. W. (2011). Return and volatility transmission in U.S. housing markets. *Real Estate Economics*, 39(4):701–741.

Office of Federal Housing Enterprise Oversight (2004). Mortgage markets and the enterprises in 2003. Research paper, Office of Federal Housing Enterprise Oversight.

Paciorek, A. (2013). Supply constraints and housing market dynamics. *Journal of Urban Economics*, 77:11–26.

Payne, J. E. and Sun, X. (2023). Time-varying connectedness of metropolitan housing markets. *Real Estate Economics*, 51:470–502.

Pesaran, H. H. and Shin, Y. (1998). Generalized impulse response analysis in linear multivariate models. *Economics Letters*, 58:17–29.

Prüser, J. and Schmidt, T. (2021). Regional composition of national house price cycles in the US. *Regional Science and Urban Economics*, 87:103645.

Racine, J. (2000). Consistent cross-validatory model-selection for dependent data: *hv*-block cross-validation. *Journal of Econometrics*, 99(1):39–61.

Roodman, D., MacKinnon, J. G., Nielsen, M. Ø., and Webb, M. D. (2019). Fast and wild: Bootstrap inference in Stata using boottest. *Stata Journal*, 19(1):4–60.

Roy Chowdhury, S., Gupta, K., and Tzeremes, P. (2023). US housing prices and the transmission mechanism of connectedness. *Finance Research Letters*, 58:104636.

Saiz, A. (2010). The geographic determinants of housing supply. *The Quarterly Journal of Economics*, 125(3):1253–1296.

Schubert, G. (2024). House price contagion and U.S. city migration networks. Technical Report 4984580, SSRN.

Sinai, T. and Souleles, N. S. (2013). Can owning a home hedge the risk of moving? *American Economic Journal: Economic Policy*, 5(2):282–312.

Song, J. (2025). Geography of corporate networks and housing price spillovers: evidence from U.S. states. *The North American Journal of Economics and Finance*, 80:102490.

Tidwell, A., Lu, Y., Lee, J., and Banerjee, P. (2023). Nature of comovements in US state and MSA housing prices. *Real Estate Economics*, 51(4):959–989.

Tsai, I.-C. and Lin, C.-C. (2019). Variations and influences of connectedness among US housing markets. *International Real Estate Review*, 22(1):27–58.

U.S. Bureau of Economic Analysis (2025). Regional economic accounts: State and county personal income and employment by industry (sainc7s, sainc7n, cainc4, cainc5s, cainc5n). `https://apps.bea.gov/regional/downloadzip.htm`. Accessed September 25, 2026.

U.S. Census Bureau (2023). Core based statistical areas (cbsas), metropolitan divisions, and combined statistical areas (csas), july 2023 delineation file. `https://www.census.gov/geographies/reference-files/time-series/demo/metro-micro/delineation-files.html`.

U.S. Department of Agriculture, Economic Research Service (1999). Natural amenities scale. `https://www.ers.usda.gov/data-products/natural-amenities-scale`. Accessed September 25, 2026.

Vansteenkiste, I. (2007). Regional housing market spillovers in the US: Lessons from regional divergences in a common monetary policy setting. Working Paper 708, European Central Bank.

Webb, M. D. (2023). Reworking wild bootstrap-based inference for clustered errors. *Canadian Journal of Economics*, 56(3):839–858.

Wu, Y.-L., Lu, C.-L., Chen, M.-C., and Chu, F.-N. (2017). What forces drive the dynamic interaction between regional housing prices? *International Journal of Strategic Property Management*, 21(3):225–239.

Yang, J., Yu, Z., and Deng, Y. (2018). Housing price spillovers in China: A high-dimensional generalized VAR approach. *Regional Science and Urban Economics*, 68:98–114.

Yurtoğlu, O. (2026). How should a penalized VAR be tuned to measure connectedness? Working paper, Koç University.

Zhu, B., Füss, R., and Rottke, N. B. (2013). Spatial linkages in returns and volatilities among U.S. regional housing markets. *Real Estate Economics*, 41(1):29–64.

Zou, H. and Hastie, T. (2005). Regularization and variable selection via the elastic net. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 67(2):301–320.

A Data Construction

A.1 House-price growth

Table A.1 reports monthly growth in the 48-state sample. Average appreciation and volatility differ across states, with many sample extrema occurring during the 1981–82 recession. California and Washington exhibit particularly large monthly declines, of about 10 percent in a month, and Minnesota of about 7 percent. In California the fall is followed within a year by a rise of similar size, and Washington's largest monthly rise is about 6 percent.[50]

[50]These months fall at the peak of mortgage rates—the 30-year fixed rate averaged 18.5 percent in October 1981 (Freddie Mac, 2026b)—when conventionally financed resales nearly stopped. California sales collapsed in 1981 while seller-financed transactions disguised price declines (Case and Shiller, 1987). The FMHPI pairs only sales financed by loans that Freddie Mac or Fannie Mae purchased, so in 1981–82 it rests on a thin and selected set of transactions, and repeat-sales indexes are noisy in thin samples (Goetzmann, 1992). The months are retained in the benchmark, as published. Re-estimating the eight annual networks whose windows contain them, 1985–1992, with each state's monthly growth winsorized at its 1st and 99th percentiles leaves the raw receiving-side coefficient negative in each of 1985 through 1988 (Online Appendix Section OA.4.2).

Table A.1: Monthly state house-price log growth, raw and with the national factor removed

State	Mean	SD	Min	Max	State	Mean	SD	Min	Max
Panel A. Raw growth, February 1975–December 2025, 611 months									
AL	0.317	0.450	-1.085	1.980	NC	0.381	0.488	-1.511	2.600
AR	0.300	0.538	-3.510	3.145	ND	0.336	0.498	-2.375	2.699
AZ	0.406	1.006	-4.360	4.448	NE	0.344	0.411	-0.707	1.648
CA	0.497	1.192	-10.256	10.128	NH	0.420	0.660	-1.470	2.229
CO	0.441	0.575	-1.268	2.209	NJ	0.429	0.672	-1.850	2.588
CT	0.378	0.682	-1.248	3.461	NM	0.369	0.533	-2.537	2.743
DE	0.390	0.507	-0.848	1.672	NV	0.406	0.973	-3.508	3.898
FL	0.408	1.078	-4.802	6.346	NY	0.418	0.640	-2.860	4.839
GA	0.356	0.551	-1.690	2.109	OH	0.329	0.594	-4.910	4.013
IA	0.319	0.403	-1.484	1.744	OK	0.317	0.516	-1.585	2.300
ID	0.412	0.722	-1.530	3.304	OR	0.458	0.729	-1.709	2.567
IL	0.345	0.718	-4.137	4.503	PA	0.373	0.509	-1.684	2.788
IN	0.328	0.547	-4.140	3.492	RI	0.458	0.747	-1.623	3.593
KS	0.327	0.458	-1.781	1.847	SC	0.365	0.536	-1.477	3.389
KY	0.335	0.371	-0.713	1.805	SD	0.341	0.461	-1.340	1.901
LA	0.313	0.473	-1.636	2.047	TN	0.370	0.535	-2.312	2.638
MA	0.475	0.651	-1.181	2.493	TX	0.359	0.771	-3.731	5.909
MD	0.384	0.676	-2.117	4.104	UT	0.435	0.647	-1.355	2.783
ME	0.422	0.633	-1.398	2.377	VA	0.396	0.532	-2.109	1.838
MI	0.349	0.739	-4.278	5.776	VT	0.398	0.533	-1.107	2.246
MN	0.362	0.868	-7.047	7.480	WA	0.498	1.040	-10.303	6.321
MO	0.346	0.697	-3.482	4.558	WI	0.367	0.573	-4.328	3.167
MS	0.275	0.554	-3.299	2.119	WV	0.281	0.720	-3.505	3.342
MT	0.427	0.632	-2.180	2.772	WY	0.376	0.594	-1.873	2.264
Panel B. First principal component removed, January 1976–December 2025, 600 months									
AL	0.007	0.313	-1.361	1.503	NC	0.003	0.319	-1.685	1.565
AR	0.005	0.333	-1.198	1.497	ND	0.030	0.396	-1.766	2.124
AZ	0.040	0.569	-3.100	2.242	NE	-0.006	0.328	-0.941	1.148
CA	0.021	0.869	-8.815	7.668	NH	0.000	0.456	-1.390	1.533
CO	-0.055	0.399	-1.585	2.421	NJ	0.024	0.457	-2.159	1.683
CT	0.029	0.480	-1.830	2.574	NM	-0.011	0.373	-2.290	2.619
DE	0.001	0.326	-1.022	1.039	NV	-0.008	0.526	-1.606	2.448
FL	0.034	0.684	-3.993	5.061	NY	0.004	0.492	-3.115	3.936
GA	-0.024	0.329	-1.773	1.663	OH	0.041	0.467	-4.158	3.736
IA	0.023	0.299	-0.979	1.292	OK	-0.032	0.403	-1.508	1.524
ID	-0.033	0.439	-1.270	1.493	OR	-0.005	0.493	-1.437	2.360
IL	0.028	0.603	-4.491	4.409	PA	0.018	0.352	-2.203	1.956
IN	0.057	0.390	-3.011	1.875	RI	0.044	0.482	-1.382	2.741
KS	0.014	0.337	-1.738	1.766	SC	0.000	0.389	-2.131	2.587
KY	0.014	0.257	-0.905	0.936	SD	-0.014	0.355	-1.264	1.416
LA	-0.027	0.363	-1.623	1.101	TN	0.018	0.378	-2.176	2.274
MA	-0.046	0.468	-1.274	1.381	TX	-0.045	0.629	-3.948	5.887
MD	0.028	0.422	-1.484	3.087	UT	-0.020	0.381	-1.009	1.239
ME	0.035	0.459	-1.976	1.499	VA	0.028	0.291	-1.793	1.105
MI	0.067	0.592	-5.462	4.765	VT	0.045	0.355	-0.820	1.476
MN	0.016	0.692	-5.523	7.150	WA	-0.016	0.841	-8.795	7.884
MO	0.005	0.523	-3.073	3.110	WI	0.024	0.463	-4.103	3.647
MS	-0.009	0.492	-3.516	2.071	WV	0.073	0.574	-1.893	2.858
MT	-0.006	0.406	-1.523	1.261	WY	-0.026	0.444	-1.412	1.510

Note: The table reports $100\Delta \log P_{i,t}$ in percentage points. Each state has 611 monthly growth observations from February 1975 through December 2025. The VAR uses the corresponding log-growth series without the factor of 100. Alaska and Hawaii are excluded. The purged panel reports the growth rates on which the factor-adjusted network is estimated (Appendix C). For each annual network's window, these are the residuals from regressing each state's growth on the window's first principal component, kept for the window's final twelve months, and in the first window for all of its months.

B VAR and GFEVD Diagnostics

B.1 Estimator choices and the fitted benchmark

Three choices of the benchmark are not neutral for what the network measures. The lag order is the one the Bayesian information criterion (BIC) supports. Computed on the benchmark estimator itself at the 41 annual networks, with the penalized fit's nonzero coefficients as degrees of freedom, BIC rejects one lag by 857 and three lags by 430 on average, and two lags has the lowest criterion in 40 of the 41 years (Online Appendix Table OA.1). With each equation refitted by least squares on its retained slopes the criterion does not separate two lags from three, and the shorter is taken. Two lags is also the order that the information criteria select in the connectedness applications this design follows, the original Diebold–Yilmaz spillover index (Diebold and Yilmaz, 2009) and the regional housing system of Antonakakis et al. (2018). The 120-month window is not varied. It also sets the ten years over which the first principal component is removed and the labor-demand shock is predicted, and the migration pools end before it begins, so a different window would change those objects as well as the network. The penalty is selected at the minimum of validation loss rather than by the one-standard-error rule, because the one-standard-error rule retains only half to two-thirds as many cross-state slopes and the cross-state lag structure is what the measurement exists to recover. The own lags carry the cross-lag penalty for the same reason, since exempting them thins the cross-state block. None of these choices optimizes residual whiteness, which is not a selection criterion here.

Three diagnostics describe how the fitted system behaves at the 41 annual networks. It is dynamically stable there, with a largest companion-matrix radius of 0.978. The penalty floor binds for two of the 1,968 annual-network equations, and no equation sits at the upper end of the path, where every coefficient would be shrunk to zero. And the residuals remain serially dependent at this lag order. Their median absolute lag-one autocorrelation is 0.39 in sample, against 0.86 for monthly price growth itself, and 0.60 on the validation blocks that did not train on each month, a figure raised partly by how the held-out series is assembled, from twelve blocks predicted by twelve fold models, each with its own offset. Two qualifications limit the reading of the Ljung–Box rejection rates that accompany the estimator. These are residuals after equation-specific elastic-net selection and shrinkage, for which the conventional reference distribution is not calibrated (Zou and Hastie, 2005), and the published FMHPI series are seasonally adjusted with two-sided filters that can themselves carry dependence across adjacent observations (Karamon, 2023; Findley et al., 1998). The rejection share of 100 percent of equations on the benchmark's held-out residuals is therefore a descriptive warning rather than a model-failure rate. Freeing the own lags from the penalty, and extending them to eight, weakens the dependence considerably, and the price is stated rather than paid. The eight-own-lag system has no stable variance decomposition in nine of the 41 annual networks, 1985 and 2002 through 2009, and it retains 5.7 active cross-state slopes per equation against the benchmark's 23.5. Section 6.1 reports both sides of that exchange, and Online Appendix Section OA.3.1 the stability and residual diagnostics of the three networks.

B.2 Validation design and penalty path

The benchmark estimates 48 equations with 96 lagged slopes in a VAR(2) and 120 response months. The ordinary elastic net has a mixing weight of 0.5, a 100-point penalty path with lower ratio 10^{-4}, and hv-block cross-validation (Burman et al., 1994; Racine, 2000). The 120 response months are cut into twelve contiguous blocks of ten, each block is held out in turn, and the months on either side of it are dropped from the training sample, as many as the maximum lag order of the equation, two for the benchmark and the own-lags-free network and eight for the eight-own-lag network. Plain K-fold validation is valid for autoregressions only when the errors are uncorrelated (Bergmeir et al., 2018); with serially correlated residuals, the buffer keeps the months adjacent to a validation block from training on the observations they are then asked to predict. The penalty is chosen equation by equation at the minimum of the cross-validated error, and own lags carry the same penalty as cross-state lags. The two own-lag alternatives are carried beside the benchmark for the state directional and allocation results. The metropolitan results use the benchmark alone, because the metropolitan system with its own lags exempt from the penalty is unstable in many windows (Online Appendix Section OA.3.4).

Which own-lag treatment is the benchmark matters for the level of the index and for the receiving side. On the 16 years from 2010 in which all three networks are stable, mean systemwide connectedness is 87.4 percent under the benchmark, 83.0 with the own lags free and 72.1 with eight own lags, and in the annual networks from 2005 the networks thin from 25.4 active cross-state slopes per equation to 13.2 and 6.7. Stability does not extend to every window of the monthly run. The benchmark companion matrix has an eigenvalue outside the unit circle in six of the 486 monthly windows, March through June 2004 and March and April 2005, the own-lags-free network at twelve, in the same two springs and in 2008, and the eight-own-lag network at 110 of its 484 monthly windows, concentrated in 2002–2012. A companion matrix with an eigenvalue outside the unit circle has no convergent moving-average representation and therefore no variance decomposition, so those windows are flagged and left blank in the monthly exhibits rather than patched with a penalty guard, and the eight-own-lag second stage is estimated on the 32 raw and 38 factor-adjusted annual networks for which that system exists.

C Factor-Adjusted Networks

C.1 Construction

The factor-adjusted networks of Section 2.3 apply the benchmark estimator to growth rates from which each window's first principal component has been removed. For a window t, let Y be the $120 \times N$ matrix of monthly log growth in the response window. Each column is standardized with the window's own mean and standard deviation, the first right singular vector r of the standardized matrix gives the factor score $f_s = z_s' r$ for month s, signed so that it correlates positively with the cross-state mean of growth, and each state's growth is regressed on f_s and a constant over the 120 response months. The residuals are the factor-adjusted series; the lag rows before the window are projected with the same means, standard deviations, singular vector and coefficients. Inside every hv-block fold the four steps are repeated on the fold's training months alone, and the validation block, the buffer months and the lag rows are projected through the fold's own purge, so no validation month informs the purge used to score it. The final fit uses the window-level purge. The metropolitan system purges its own first component across its 100 areas. For each state, 100 times the R^2 of its growth on the window's factor score is the share of its growth variance on the first principal component used in Section 2.3, in Figure C.1 and in Online Appendix Table OA.3.

The national factor has come to dominate state house-price growth. Averaged across states and windows, the first component accounts for 46 percent of a state's growth variance. The share rises from 22 percent in the windows ending in 1985–1989 to 27 percent in the 1990s, 38 percent in the 2000s, 67 percent in the 2010s and 69 percent since 2020, and in every window it differs widely across states (Figure C.1).

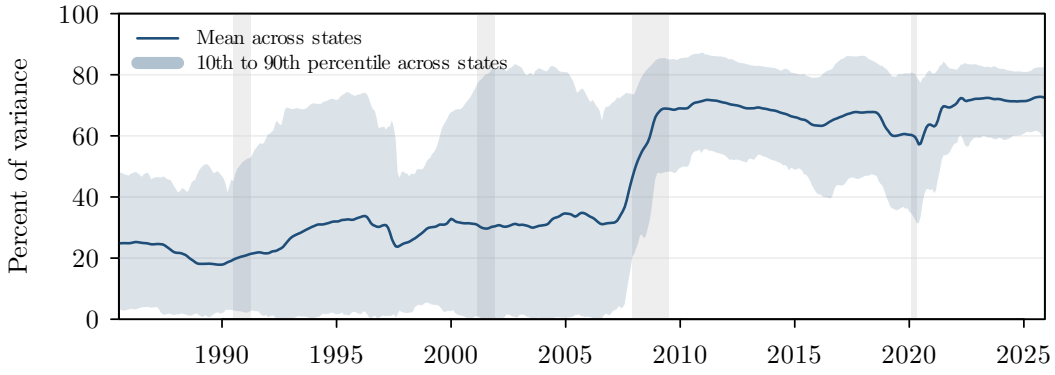


Figure C.1: First-component share of state growth variance, July 1985–December 2025

Note: In every monthly window, the share of each state's monthly growth variance accounted for by the first principal component of the window's standardized 48-state growth panel. The line is the mean across states, which for standardized growth is the component's share of total variance, and the band the range from the 10th to the 90th percentile across states. Shaded areas denote NBER recessions.

The rolling component is the same factor from one window to the next, its loadings correlating at 0.95 twelve months apart at the median window, and what it leaves is regional: after the purge

the mean pairwise correlation of states in the same Census region is 0.11 at the median window, against -0.04 for states in different regions. The purge lowers both the first-component share and the mean absolute pairwise correlation of the series in 39 of the 41 annual windows of each network. The two exceptions, 1997 and 2006 for the states, are windows with a weak common factor, in which the mean absolute correlation still falls. The factor-adjusted state benchmark, the network with free own lags and the metropolitan network are dynamically stable in all 41 annual networks, and the eight-own-lag network in 38. The smallest factor-adjusted external connectedness is 32.5 percentage points, so the log outcomes are well defined, and the within-year standard deviation of log external connectedness is 0.189, against 0.122 raw.